

# Rayfield-Based Calibration and Effective Physical Model Identification of a Common Main Objective Stereo Microscope

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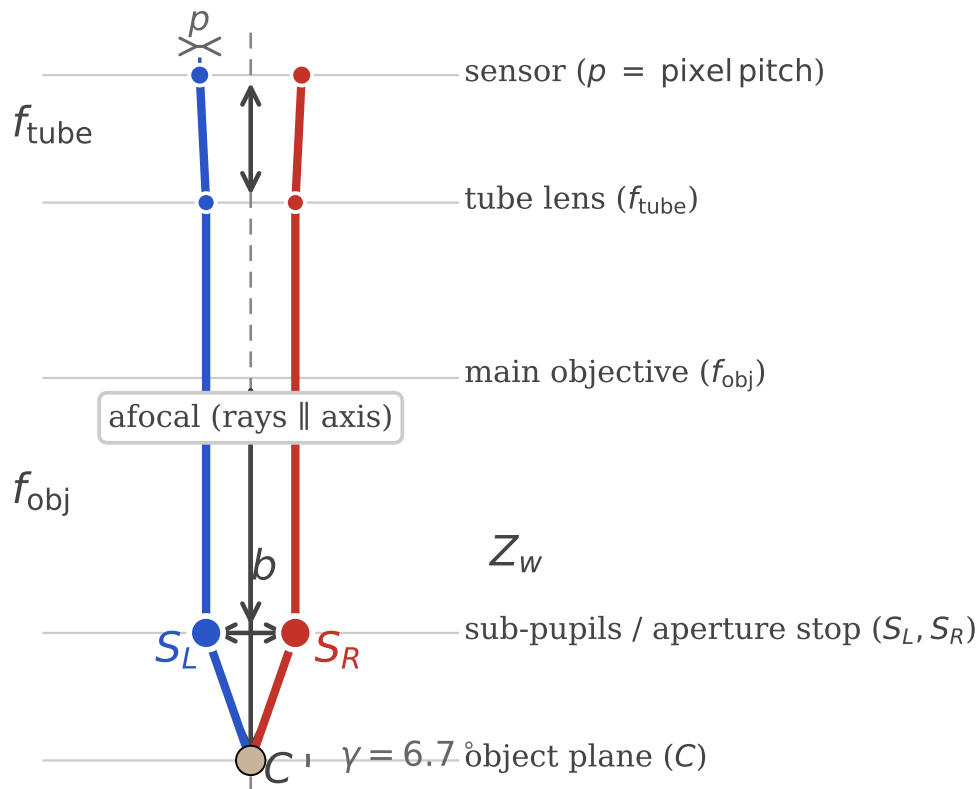
## Abstract

Common Main Objective (CMO) stereo microscopes violate the single-optical-centre assumption of standard camera calibration, making the joint estimation of optical parameters and board poses ill-conditioned: a Schur-complement diagnostic of the Fisher information gives an optics–pose coupling norm  $c = 0.98$  on a real CMO (Pycaso) configuration. We propose a two-stage approach. First, a Zernike rayfield—a per-pixel 3D line—is fitted to ChArUco corners, decoupling measurement from model identification. Second, a compact physical model is built by iterative residual analysis: the Zernike decomposition of the residual between candidate model and measured rayfield reveals which physical degree of freedom is missing. On a single CMO instrument, the rayfield-identified 26-parameter model—a quasi-telecentric CMO skeleton with per-channel SE(3) arm alignment—reaches 1.06 px reprojection error, versus  $> 300$  px for standard OpenCV calibration; used to initialise a free-pose bundle adjustment stabilised by a Schur-spectrum observability prior, the same model refines to 0.28 px, within 0.1 px of an interpretation-free Soloff polynomial floor (0.18 px), while remaining physically interpretable. External profilometry of an independent specimen resolves the residual CMO–Zernike depth-scale ambiguity ( $s_z \approx 0.70$ ): the CMO 26p and Soloff reliefs match the profilometric ground truth to  $\approx 10\%$ , whereas the dense Zernike reconstruction over-amplifies the relief by  $1.63\times$  (traceable accuracy on a certified 3D artefact remains future work). The rayfield thus acts as a diagnostic instrument—identifying the missing degree of freedom and supplying the observability prior—not merely as a calibration model.

**Keywords:** stereo microscope; CMO; non-central calibration; Zernike rayfield; BIC model selection; telecentricity

## 1 Introduction

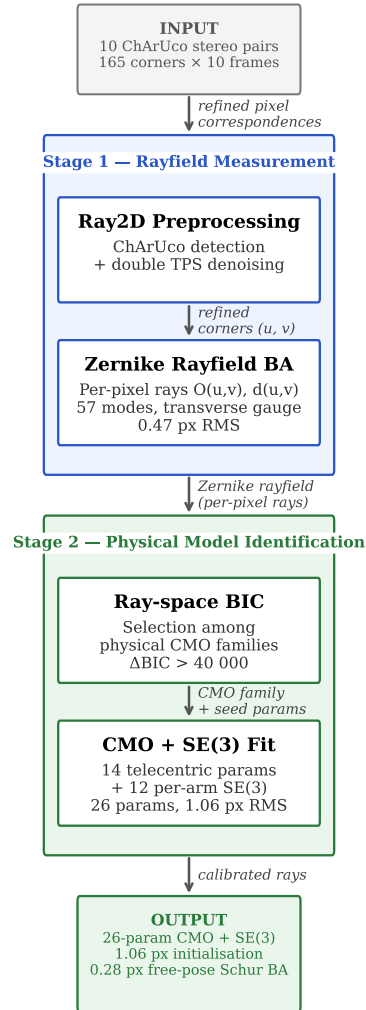
Stereo calibration is a mature technology when the camera obeys the pinhole model: all chief rays pass through a single optical centre, and mature toolchains (OpenCV, MATLAB, Bouguet) produce reliable results. This assumption breaks down for Common Main Objective (CMO) stereo microscopes, which use a single large objective shared between two off-axis sub-apertures [11, 23]. Each channel’s chief rays appear to originate from a different effective sub-pupil, separated by the stereo baseline (Figure 1). OpenCV’s `stereoCalibrate` [18] assumes one optical centre per camera; on the CMO architecture it produces unusable results ( $> 300$  px reprojection error under the tested configuration).



**Figure 1.** CMO stereo microscope optical layout. Two off-axis sub-pupils share a single main objective. Chief rays converge toward the object plane through different effective origins—not through a single common centre.

### Two-stage decomposition: measure rays first, identify optics second

Stage 1 measures rays (Ray2D is 2D; Zernike BA imposes only generic 3D ray consistency, no physical CMO model); Stage 2 identifies optics in ray space.



**Figure 2.** Pipeline overview. The two-stage decomposition separates rayfield measurement (Ray2D preprocessing + Zernike BA) from physical model identification (ray-space BIC selection). The Ray2D stage is purely 2D and does not impose a 3D camera model.

The broader problem is that many optical instruments—CMO and Greenough stereo microscopes, endoscopes, Scheimpflug cameras, multi-camera arrays—depart from the central projection model. Among these, the CMO architecture is particularly challenging: in a Greenough microscope the two independent objectives yield two well-separated optical centres that can be modelled as individual perspective cameras, whereas a CMO microscope shares a *single* large objective between off-axis sub-apertures, producing decentred exit pupils whose chief rays do not converge to a common point—a subtle geometric coupling that violates the pinhole assumption far more severely than a simple dual-perspective system. Generic camera models [10, 30, 25] and recent stereo microscope calibration studies [26, 35, 14] highlight the need for non-central approaches, but do not provide a systematic method for identifying the correct physical model family from observations (see Section 2 for a detailed review).

StereoComplex [33] addresses this gap: it fits a Zernike rayfield  $\mathcal{R}(u, v) = (O(u, v), d(u, v))$ —a 3D line for every pixel—to corner observations, using a Zernike polynomial basis [34, 17]. This two-stage decomposition—measure rays first, identify optics second (Figure 2)—is the central insight: it decouples the severely ill-conditioned joint estimation of optical parameters and board poses into a well-conditioned rayfield fit followed by pose-free model selection [25]. From this *measured* rayfield, physical descriptors can be read directly (Section 3), structural mismatches diagnosed by residual modal analysis (Section 3.3), and compact physical models constructed iteratively (Section 3.4–3.5).

This paper presents the first real-data validation of the StereoComplex rayfield-mediated non-central calibration workflow on a CMO stereo microscope using legacy ChArUco [9, 22] images from the open Pycaso dataset [8]. The contributions are:

1. A complete pipeline from raw ChArUco images to a compact 26-parameter physical CMO model, validated on real data. The model provides *effective* identification: the shear coefficients and telecentric skeleton are data-supported, while geometric scale parameters (WD,  $b$ ,  $f_{\text{obj}}$ ) and most SE(3) components are absorbed by the rayfield gauge and pose coupling (see §3.7 for a per-parameter identifiability analysis).
2. A residual analysis method—Zernike modal decomposition of direction and moment differences between a candidate physical model and the measured rayfield—that identifies missing degrees of freedom.
3. A two-level model-selection framework: ray-space BIC is used to identify the optical family, while a separate operational reprojection guard rejects models whose pixel RMS is not usable for calibration.
4. Demonstration that the Zernike rayfield serves a *double role* in direct bundle adjustment: (i) as an initialiser, it places the BA in the correct convergence basin, improving pixel RMS from 1.06 px to 0.88 px (17 %) while leaving all 26 model parameters unchanged to within  $10^{-3}$ ; (ii) as an *observability prior*, the Schur complement of the Fisher information matrix at the rayfield solution defines an anisotropic penalty that suppresses optics–pose compensation—the prior reduces weak-mode parameter drift by a factor of  $280\times$  (from 0.93 to 0.0033) at a cost of only 0.038 px in 2D reprojection RMS. In the free-pose Schur-stabilised BA, the 26-parameter physical model reaches 0.28 px—within 0.1 px of the Soloff polynomial floor—with its optics held in the observable

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subspace rather than drifting into the unobservable directions that the unregularised optimum (0.24 px) exploits.

## 2 Related Work

### 2.1 Generic and Non-Central Camera Calibration

The pinhole camera model with polynomial distortion [36, 5] remains the standard in computer vision toolkits [18, 4]. For wide-angle and fisheye lenses, unified models [15, 12, 24] extend the pinhole paradigm with additional radial terms. However, these parametric models share a fundamental assumption: all chief rays pass through a single optical centre.

For cameras that violate this assumption, generic (non-parametric) models assign a per-pixel ray without imposing a central projection constraint. Grossberg and Nayar [10] introduced the raxel model, treating each pixel as an independent ray. Sturm and Ramalingam [30], later refined by Ramalingam and Sturm [20], developed a generic calibration framework from multiple views of a planar target, showing that a flexible pixel-to-ray mapping can be estimated without knowing the camera model *a priori*. Miraldo and Araujo [16] proposed smooth generic camera models using continuous ray-field interpolation. Soloff et al. [28] introduced polynomial expansions mapping 3D coordinates to 2D pixel positions; this approach, widely adopted in stereo-microscope calibration [8], treats the camera as a black-box regression problem and can achieve excellent pixel accuracy—as we confirm in §4.6, a degree-3 Soloff fit reaches 0.18 px RMS on the Pycaso dataset with 80 parameters. Camposeco et al. [7] demonstrated spline-based generic calibration with subpixel accuracy. Ramalingam, Sturm, and Nayar [21] unified these approaches in a generic structure-from-motion framework.

Most recently, Schöps et al. [25] showed that highly parameterised generic models (10,000+ parameters) can outperform classical 12-parameter models on standard benchmarks, challenging the assumption that parametric models are inherently more robust.

Our work differs from these generic approaches in two ways: (1) we use the Zernike rayfield not as a final calibration model but as a *diagnostic instrument* to discover the correct physical model family, and (2) we construct a compact, interpretable physical model guided by residual structure rather than imposing a generic non-parametric fit. Importantly, we do not claim superior pixel accuracy—the Soloff polynomial is more accurate at equivalent parameter counts (§4.6)—but superior physical interpretability: unlike a black-box polynomial, the 26-parameter model has a physically interpretable parameterisation (baseline, working distance, convergence angle, shear, SE(3) arm poses), although only some combinations are independently identifiable after pose marginalisation (§3.7). Interpretability is, moreover, not the rayfield’s only advantage over a polynomial regressor: although the Soloff fit achieves lower *in-sample* pixel RMS, the rayfield’s 3D ray constraint acts as a *structural regulariser*. On held-out extreme poses the degree-3 Soloff fit degrades by  $4.65\times$  while the 180-parameter Zernike rayfield degrades by only  $1.86\times$  (§4.6), because each pixel’s ray must stay geometrically consistent with a rigid 3D board rather than memorising individual training poses. The residual accuracy gap is, moreover, small in absolute terms: once the rayfield-identified 26-parameter model is refined by the free-pose Schur-stabilised bundle adjustment (§3.9), its reprojection RMS falls from the 1.06 px initialisation to 0.28 px—within 0.1 px of the Soloff floor (0.18 px)—so physical interpretability costs a fraction of a pixel, not a category of accuracy.

## 2.2 Stereo Microscope Calibration

Stereo microscopes pose a specific calibration challenge: the two optical channels share a common main objective but have separate effective pupils [26]. Digital image correlation under optical microscopes requires careful distortion correction [35, 14].

These methods calibrate a specific instrument but do not address the generic problem of *identifying* the correct optical model family from observations. Our contribution is a systematic framework that measures the ray-field first, then uses residual analysis to construct a physical model that matches the observed structure—here, discovering the telecentric CMO family and the SE(3) arm misalignment.

## 2.3 Model Selection in Camera Calibration

Model selection in calibration traditionally relies on reprojection RMS, which penalises underfitting but is insensitive to overfitting. The Akaike Information Criterion (AIC) [1] and Bayesian Information Criterion (BIC) [27] add parameter-count penalties, providing a principled trade-off between fit quality and model complexity. Strecha et al. [29] established benchmark datasets and evaluation protocols for camera calibration and multi-view stereo. Our work extends BIC-based model selection to ray space: we compare optical model families (pinhole, Brown-Conrady, parallel-plate, telecentric CMO) against a measured Zernike rayfield, and add an operational reprojection guard that rejects models whose pixel RMS exceeds a usability threshold—a constraint that the raw ray-space BIC does not capture.

## 2.4 Plücker Coordinates and Line-Based Geometry

Plücker coordinates [19] represent a 3D line as a 6-vector  $(d, m)$  where  $d$  is the direction and  $m = O \times d$  is the moment. They are widely used in geometric computer vision for line-based structure from motion and calibration [2, 31]. Our residual analysis uses Plücker moments to separate direction errors from origin errors:  $\Delta d$  reveals angular misalignment;  $\Delta m$  reveals the effective ray origin shift. The  $Z_0^0$ -dominated residual in both  $\Delta d$  and  $\Delta m$  is the key diagnostic that guides the SE(3) arm alignment.

## 2.5 Residual-Guided Model Construction

The idea of using residuals to guide model refinement has roots in adaptive optics and wavefront sensing [34, 17], where Zernike decomposition of a distorted wavefront identifies specific aberrations (defocus, astigmatism, coma) that are then corrected by deformable mirrors. Our method applies the same principle to camera calibration: the Zernike decomposition of the residual between a candidate physical model and the measured rayfield identifies which physical degree of freedom is missing—a  $Z_0^0$ -dominated residual points to a global offset (SE(3) arm misalignment); a  $Z_2$ -dominated residual would point to astigmatism or field curvature. This is distinct from generic non-parametric approaches [25] that fit a flexible model without extracting physical meaning from the residual structure.

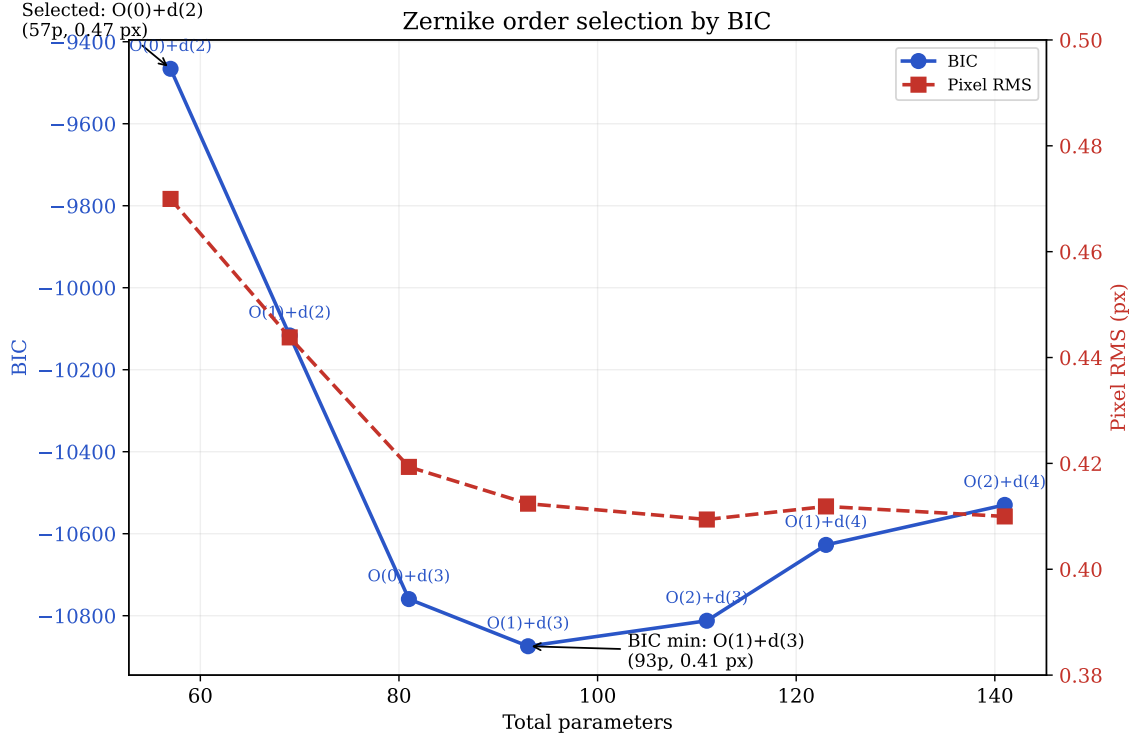
# 3 Method

## 3.1 Zernike Order Selection by BIC

The Zernike rayfield has two hyperparameters: the origin polynomial order  $o$  and the direction polynomial order  $d$ . Increasing either adds Zernike modes, reducing pixel RMS at the cost of more parameters. We sweep  $o \in \{0, 1, 2\}$  and  $d \in \{2, 3, 4\}$  (7 combinations) and select the optimal order by BIC [27] (Figure 3):

$$\text{BIC}_{\text{px}} = k \log(N_{\text{obs}}) + N_{\text{obs}} \log(\text{RSS}_{\text{px}}/N_{\text{obs}}), \quad (1)$$

where  $k$  is the total number of parameters (Zernike coefficients + poses),  $N_{\text{obs}} = 3300$  corner observations, and  $\text{RSS}_{\text{px}}$  is the sum of squared pixel-equivalent residuals. The BIC treats these 3300 corners as independent Gaussian observations; corners within a frame are in fact spatially correlated, so the absolute BIC magnitudes are optimistic. The *family*-level selection used here is nonetheless robust: the gaps between optical families ( $\Delta\text{BIC} > 40\,000$ ) dwarf any plausible reduction in the effective sample size.



**Figure 3.** Zernike order selection by BIC. Pixel RMS falls from 0.47 px at 57 params to 0.41 px near the BIC minimum (93 params) and then saturates within corner-detection noise (it is not strictly monotonic beyond, varying by  $< 0.005$  px up to 141 params), but BIC penalises complexity. The BIC minimum is at O(1)+d(3) (93 params, BIC =  $-5093$ , RMS = 0.41 px), though O(0)+d(2) (57 params, BIC =  $-4522$ ) is within  $\sim 10\%$  in RMS at 40 % fewer parameters. The O(0)+d(2) model (57 params, 0.47 px) is selected for physical model construction because it is internally consistent with the rigid-sub-pupil CMO assumption, achieves 90 % of the noise-floor accuracy at nearly half the parameters of the BIC minimum, and the marginal improvement above d(2) falls within corner detection noise.

**Table 1.** Zernike order sweep: pixel RMS and BIC.

| Model            | $k$       | RMS (px)     | $\text{BIC}_{\text{px}}$  |
|------------------|-----------|--------------|---------------------------|
| O(0)+d(2)        | 57        | 0.470        | $-4522$                   |
| O(1)+d(2)        | 69        | 0.444        | $-4803$                   |
| O(0)+d(3)        | 81        | 0.419        | $-5080$                   |
| <b>O(1)+d(3)</b> | <b>93</b> | <b>0.412</b> | <b><math>-5093</math></b> |
| O(2)+d(3)        | 111       | 0.409        | $-4995$                   |
| O(1)+d(4)        | 123       | 0.412        | $-4858$                   |
| O(2)+d(4)        | 141       | 0.410        | $-4743$                   |

Table 1 summarises the full sweep. The BIC minimum is therefore interpreted as a diagnostic of the best flexible rayfield accuracy available in this data regime. The O(0)+d(2) rayfield is not claimed to be the statistical BIC optimum; it is deliberately retained as the physically consistent reference for CMO model construction because it enforces rigid effective sub-pupils while remaining within 0.06 px of the BIC-minimum RMS.

The BIC minimum at O(1)+d(3) achieves 0.41 px RMS with 93 parameters. We nevertheless select O(0)+d(2) (57 params, 0.47 px) as the preferred rayfield configuration, for three reasons:

1. **Simplicity:** 57 params vs. 93 — nearly half the parameters for a penalty of only 0.06 px in RMS ( $\sim 0.1$  px in practice).
2. **Marginal improvement:** the residual gain from d(2) to d(3) (0.47 px  $\rightarrow$  0.41 px) is well within typical subpixel noise of ChArUco detection, so O(0)+d(2) is retained as the physically consistent reference even though O(1)+d(3) is the statistical BIC minimum.
3. **Consistency with the physical model:** the CMO physical model assumes rigid sub-pupils ( $o = 0$ ). Using the same origin order in the Zernike rayfield makes the two representations internally consistent and comparable: the Zernike field measures what the physical model can represent.

Selecting O(0)+d(2) achieves 90 % of the noise-floor accuracy with a representation that is internally consistent with the physical model it validates.

### 3.2 Per-Pixel Ray-Field and Transverse Gauge

For each pixel  $(u, v)$  on the sensor, the imaging system defines a 3D line in space. We represent this line  $\mathcal{L}_c(u, v)$  by an origin point  $O_c(u, v) \in \mathbb{R}^3$  and a unit direction vector  $d_c(u, v) \in \mathbb{S}^2$ , where  $c \in \{L, R\}$  denotes the left or right channel. Points on the line are parameterised as  $O_c + t d_c$  for  $t \in \mathbb{R}$ .

The origin  $O_c(u, v)$  is not unique: any point along the ray would represent the same line. We fix this gauge freedom by enforcing the **transverse gauge**  $O_c \cdot d_c = 0$ , which selects the point on the ray closest to the coordinate origin. This eliminates one redundant degree of freedom per ray and stabilises the parameterisation.

The origin and direction fields are expanded on a Zernike polynomial basis [34, 17]:

$$O_c(u, v) = \sum_m a_{c,m}^O Z_m(\tilde{u}, \tilde{v}), \quad d_c(u, v) = \text{normalise} \left( d_c^{\text{pinhole}}(u, v) + \sum_{m \leq d_{\max}} a_{c,m}^d Z_m(\tilde{u}, \tilde{v}) \right), \quad (2)$$

where  $Z_m$  are the real-valued Zernike polynomials on the unit disk,  $\tilde{u}, \tilde{v}$  are normalised pixel coordinates mapping the sensor to  $[-1, 1]^2$ ,  $d_c^{\text{pinhole}}$  is the initial pinhole direction from a fixed central projection matrix  $K$ , and the direction perturbation is projected transverse to  $d_c^{\text{pinhole}}$  before renormalisation. The expansion order for the origin field is  $o_{\max}$  and for the direction field is  $d_{\max}$ ; selecting  $o_{\max} = 0$  and  $d_{\max} = 2$  (see §3.1) gives 3 origin coefficients and 18 direction coefficients per channel. The Zernike polynomials here follow the Noll [17] single-index ordering on the unit disk, with the standard OSA/ANSI normalisation  $\iint_{\text{disk}} [Z_m]^2 dA = 1$  for unit-area disk; the code uses the same convention [33].



### 3.3 Ray-Space Bundle Adjustment

The rayfield coefficients  $a_{c,m}^O, a_{c,m}^d$  and the board poses  $\{\eta_p\}_{p=1}^P$  are estimated jointly from  $N$  ChArUco corner observations. Each observation  $k$  links a pixel  $(u_k, v_k)$  on channel  $c_k$  to a known 3D board point  $X_k$  expressed in the board frame. The board pose  $\eta_p = (R_p, t_p) \in \text{SE}(3)$  transforms  $X_k$  to the camera frame:  $X_k^{\text{cam}} = R_p X_k + t_p$ .

The **ray-space residual** measures the perpendicular distance from the transformed board point to the ray  $\mathcal{L}_{c_k}(u_k, v_k)$ :

$$r_k = \|(X_k^{\text{cam}} - O_{c_k}) - ((X_k^{\text{cam}} - O_{c_k}) \cdot d_{c_k}) d_{c_k}\|. \quad (3)$$

This is the norm of the component of  $X_k^{\text{cam}} - O_{c_k}$  orthogonal to the ray direction  $d_{c_k}$ . The objective is the sum of squared residuals:

$$\min_{\{a_{c,m}\}, \{\eta_p\}} \sum_k r_k^2, \quad (4)$$

solved by nonlinear least-squares.

For the CMO microscope, we impose the physically-motivated constraint that the board is mounted on a Z-only translation stage: all  $P$  frames share a common rotation  $R$  and XY translation  $(t_x, t_y)$ , with only the Z component varying per frame. This reduces the pose parameters from  $6P$  to  $3 + 2 + P = 15$  (for  $P = 10$  frames:  $60 \rightarrow 15$ ), making the 57-parameter Zernike BA well-conditioned.

The ray-space BA differs from classical reprojection minimisation [36] in a fundamental way: it optimises the 3D geometric consistency of rays rather than the 2D reprojection error. This is the measurement stage of the two-stage decomposition (see Section 4.2).

### 3.4 The Zernike Rayfield as Observable

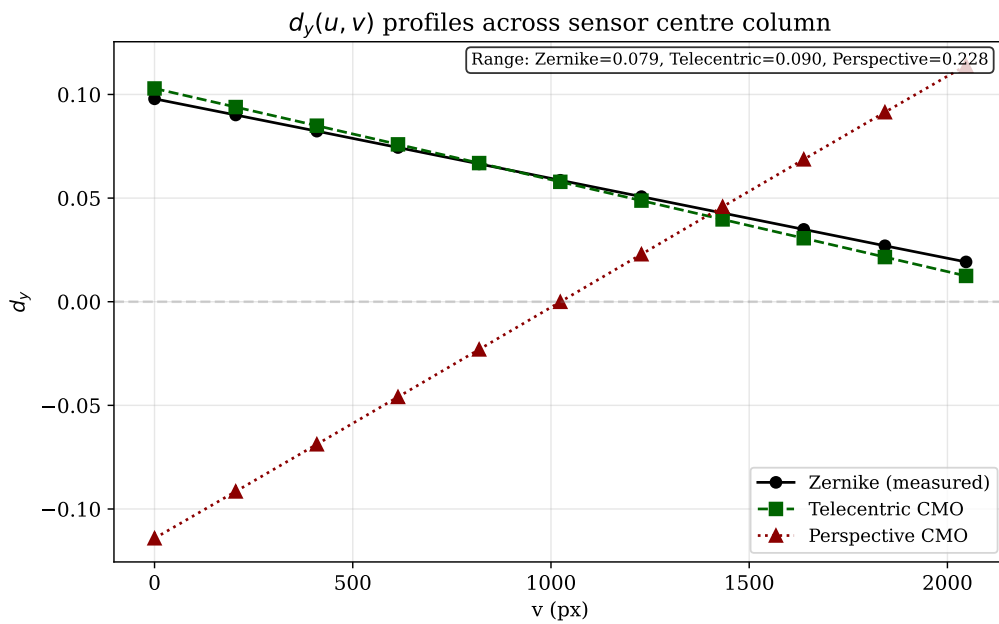
The fitted rayfield  $\mathcal{R}_c(u, v) = (O_c(u, v), d_c(u, v))$  is used as an experimental observable. Because the Zernike rayfield is a non-central model, the BA that fits it carries a scale degeneracy: the pixel projection is invariant under simultaneous scaling of all lengths and the effective focal length  $f_x$  (see §4.7 for the linear  $f_x$ - $WD$  relationship). Three independent constraints intersect to anchor the absolute scale. (1) The ChArUco board has known square size (0.3 mm); from the observed corner spacing  $\Delta u_{\text{square}}$  we obtain the ratio  $f_x/WD = \Delta u_{\text{square}}/0.3 \text{ mm} \approx 395 \text{ px/mm}$ . (2) The motorised Z-stage that acquired the Pycaso dataset displaces the board by a known physical step  $\Delta Z_{\text{stage}} \approx 0.07 \text{ mm}$  per frame ( $\sim 10\%$  accuracy); the measured disparity shift  $\Delta d$  between successive frames gives  $f_x = \Delta d \cdot WD / \Delta Z_{\text{stage}}$ . (3) the working distance documented for the Pycaso acquisition configuration is  $\approx 65 \text{ mm}$  [8]. (The bare Planapo 1.0 $\times$  objective is specified at 61.5 mm [13]; the mounted/effective working distance of the assembled instrument differs, and that 61.5 mm datasheet value anchors a separate chief-ray consistency check in §4.5.) Solving the three constraints jointly yields  $f_x = 25\,600 \text{ px}$  and  $WD = 64.7 \text{ mm}$ , consistent at the  $\sim 1\%$  level with the documented configuration working distance. All geometric descriptors ( $b, WD, f_{\text{obj}}$ ) inherit this scale and are therefore reported as *effective* quantities in the chosen gauge; an independent external reference would be required to convert them to traceable measurements (Section 5.2). The centre-pixel values, from which the effective sub-pupils  $O_L, O_R$  are reconstructed, are:  $b = \|O_R - O_L\| = 24.9 \text{ mm}$  (effective stereo baseline in the rayfield gauge),  $z_p = (|O_{L,z}| + |O_{R,z}|)/2 = 2.5 \text{ mm}$  (sub-pupil depth),  $WD = 64.7 \text{ mm}$  (effective working distance, from mean pose Z; convention: object-plane Z minus optical-axis origin),  $f_{\text{obj}} = WD - z_p = 62.2 \text{ mm}$  (effective axial length),  $\theta = \arccos(d_L \cdot d_R) = 22.6^\circ$  (effective full convergence angle at the centre pixel).

### 3.5 Physical Model Construction by Residual Analysis

The core methodology is iterative: propose a candidate physical model, compute its residual against the Zernike rayfield, decompose the residual on Zernike modes, and use the dominant mode to identify the missing degree of freedom.

#### 3.5.1 Perspective CMO (Baseline)

The simplest CMO model places a perspective camera behind each decentred sub-pupil  $S_c = (\pm b/2, 0, WD - f_{\text{obj}})$ . This predicts  $d_y(u, v) \propto (v - c_y)$ —a strong linear gradient. The measured  $d_y$  field is nearly constant across the sensor (range 0.079, mean +0.059), while the perspective model predicts a range of 0.232—a  $3\times$  discrepancy (Figure 4). The real system is **object-space telecentric**, not perspective.



**Figure 4.** Comparison of  $d_y(u, v)$  profiles across the sensor. The Zernike rayfield (measured) shows a nearly constant  $d_y$  (range 0.079), while the perspective CMO model predicts a  $3\times$  larger linear gradient (range 0.232). The telecentric model reproduces the measured profile almost exactly.

#### 3.5.2 Telecentric CMO

The perspective model fails because the real microscope is object-space telecentric: chief rays in object space are nearly parallel to the optical axis rather than diverging from a single sub-pupil point. A perspective camera predicts  $d_y \propto (v - c_y)$ —a linear gradient from the sub-pupil—while a telecentric system emits a bundle of nearly parallel rays whose direction varies only weakly across the field, explaining the factor-3 discrepancy in Figure 4. The telecentric model directly parameterises the ray-field without deriving it from a point projection. For each channel  $c \in \{L, R\}$ , the origin is a rigid sub-pupil:

$$O_c = S_c = (\pm b/2, 0, WD - f_{\text{obj}}), \quad (5)$$

where the sign is  $+$  for right and  $-$  for left. The direction field is an affine function of normalised angular coordinates  $\tilde{u} = (u - c_x)p/f_{\text{ang}}$ ,  $\tilde{v} = (v - c_y)p/f_{\text{ang}}$ , where  $p$  is the pixel pitch,  $f_{\text{ang}}$  is the angular focal length, and  $(c_x, c_y)$  is the principal point:

$$d_c(u, v) = \text{normalise}(d_{c,0} + s_{x,c} \tilde{u} e_x + s_{y,c} \tilde{v} e_y + s_{xy,c} \tilde{u} \tilde{v} e_x + s_{yx,c} \tilde{v} \tilde{u} e_y), \quad (6)$$

where  $d_{c,0} = (\pm \sin \theta, d_y^{\text{com}}, \cos \theta)$  is the chief-ray direction, with  $\theta$  the convergence half-angle and  $d_y^{\text{com}}$  a shared vertical component.

Adding pupil shear—an affine transverse variation of the origin—completes the model:

$$O_c(u, v) = S_c + P_{d_c}^\perp(\rho_{x,c} \tilde{u} e_x + \rho_{y,c} \tilde{v} e_y), \quad (7)$$

where  $P_d^\perp v = v - (v \cdot d) d$  projects onto the plane orthogonal to  $d$ , ensuring  $O_c \cdot d_c = 0$ .

The 14 parameters are:  $f_{\text{obj}}$ ,  $WD$ ,  $b$  (skeleton),  $c_x, c_y$  (principal point),  $f_{\text{ang}}$  (angular scale),  $\theta$ ,  $d_y^{\text{com}}$  (chief-ray direction),  $s_{x,L}, s_{y,L}, s_{x,R}, s_{y,R}$  (per-channel slopes),  $\rho_x, \rho_y$  (shared pupil shear). This model achieves 0.12 mm ray-space RMS (down from 3.48 mm for perspective) and 14.6 px reprojection (down from 86 px). The dominant geometry is captured, but 14.6 px is far from the Zernike reference (0.47 px).

### 3.5.3 Residual Analysis

The residual between the telecentric model and the Zernike rayfield is computed on a  $41 \times 41$  grid:  $\Delta d = d_{\text{Zernike}} - d_{\text{CMO}}$ ,  $\Delta m = m_{\text{Zernike}} - m_{\text{CMO}}$ , where  $m = O \times d$  is the Plücker moment [19].

Projecting on Zernike modes up to order 4 reveals that **both  $\Delta d$  and  $\Delta m$  are 97–98 %  $Z_0^0$  (global piston)**. This is the crucial diagnostic: the residual is not a spatial field distortion (which would appear in higher Zernike modes) but a **global line-bundle offset**—the entire set of rays from each channel is slightly rotated or translated relative to the Zernike reference.

### 3.5.4 Testing Alternative Hypotheses

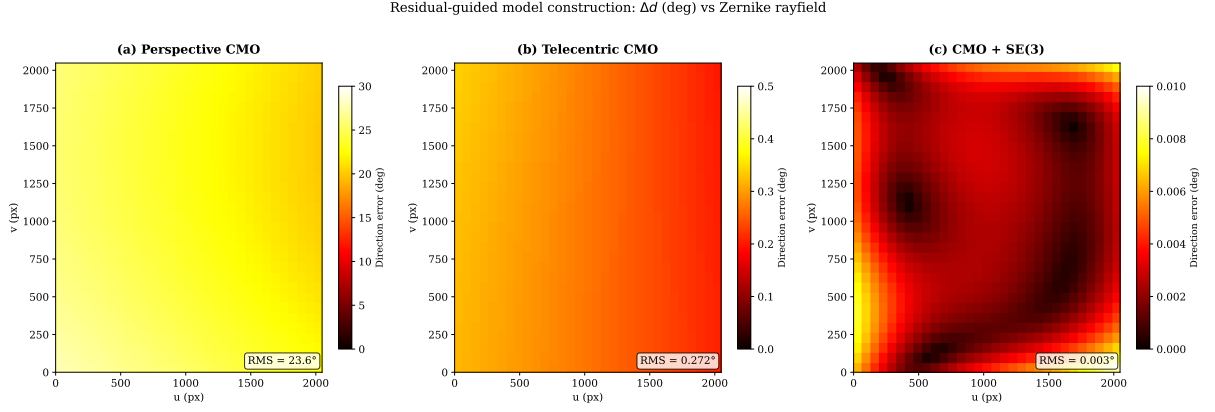
Two alternative hypotheses are tested and rejected:

- **Image-space pre-warp:** a polynomial mapping  $\xi = W(u, v)$  before the direction model. Affine and quadratic warps *degrade* pixel RMS ( $14.6 \rightarrow 16.0\text{--}16.5$  px).
- **Spatially varying origin:** affine and quadratic transverse origin fields with direction fixed. Only 8 % ray RMS improvement—the residual is not spatial.

Both negative results are consistent with the  $Z_0^0$  diagnostic (they would produce spatial, not global, changes).

### 3.5.5 Residual-Guided Hypothesis Construction

The central methodological contribution is that physical model construction is guided by *residual structure*, not blind fitting. At each stage, the direction residual  $\Delta d = d_{\text{Zernike}} - d_{\text{model}}$  is computed on a  $31 \times 31$  grid and displayed as a heatmap. The pattern of the residual directly reveals which physical degree of freedom is missing.



**Figure 5.** Residual evolution across model stages. Direction error  $\Delta d$  (degrees) between the Zernike rayfield (reference) and each candidate model, evaluated on a  $31 \times 31$  pixel grid. **Left:** Perspective CMO — large structured error (RMS  $23.6^\circ$ , max  $27.7^\circ$ ), dominated by a strong  $d_y$  gradient. **Centre:** Telecentric CMO — RMS drops to  $0.27^\circ$  (two orders of magnitude), but the residual is still  $Z_0^0$ -dominated (global piston), revealing the missing SE(3) arm alignment. **Right:** CMO + SE(3) — RMS drops to  $0.003^\circ$ , near the noise floor. The residual heatmap at each stage directly tells the user *what* is wrong and *where* to look next.

Stage 1 (Perspective CMO): the residual is large and structured, with RMS  $23.6^\circ$  (max  $27.7^\circ$ ). The  $d_y$  component shows a strong linear gradient trace—perspective projection from a point sub-pupil—that the Zernike field does not have. This **rejects** the perspective family and points to object-space telecentricity.

Stage 2 (Telecentric CMO): the residual drops to  $0.27^\circ$  RMS (two orders of magnitude), confirming that the telecentric model family is correct. However, the residual is  $Z_0^0$ -dominated (97–98 % global piston in both  $\Delta d$  and  $\Delta m$ )—a constant offset across the entire field. This **reveals** that the missing degree of freedom is a global line-bundle misalignment, not a spatial field distortion.

Stage 3 (CMO + SE(3)): the residual drops further to  $0.003^\circ$  RMS, near the noise floor. The remaining gap to the Zernike reference (0.47 px vs 1.06 px) comes from distributed low-amplitude aberrations (field curvature, astigmatism) that require the flexibility of the 57-parameter Zernike model.

The critical point is that **at no stage was a model added blindly**. The heatmap pattern at each stage dictates the next hypothesis: perspective  $\rightarrow$  telecentric (because the  $d_y$  gradient is wrong), telecentric  $\rightarrow$  SE(3) (because the residual is  $Z_0^0$  piston, not spatial).

### 3.5.6 Per-Channel SE(3) Arm Alignment

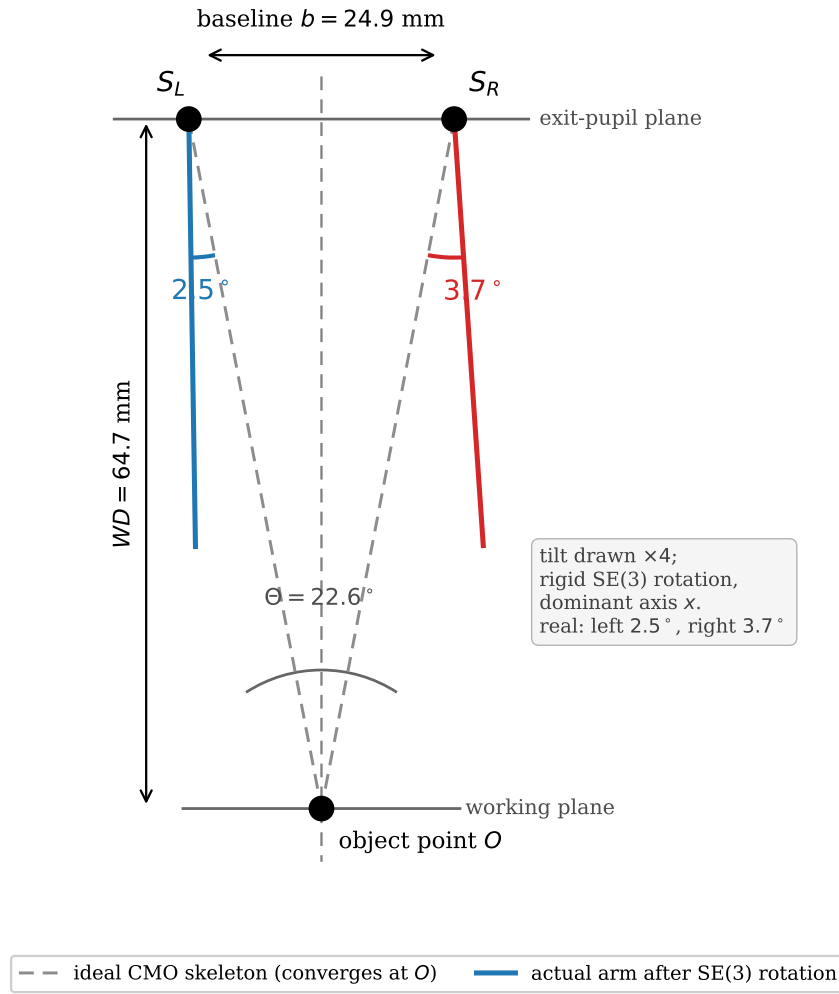
The  $Z_0^0$ -dominated residual points to a global misalignment of each optical arm. A rigid transform  $(R_c, t_c) \in \text{SE}(3)$  is applied to each channel’s ray bundle, representing a small rotation and translation of the entire optical arm relative to the ideal CMO skeleton. For each pixel  $(u, v)$  on channel  $c$ , the telecentric ray  $(O_c^{\text{tel}}, d_c^{\text{tel}})$  is transformed:

$$d_c(u, v) = R_c d_c^{\text{tel}}(u, v), \quad O_c(u, v) = R_c O_c^{\text{tel}}(u, v) + t_c. \quad (8)$$

Each  $R_c \in \text{SO}(3)$  is parameterised by a 3-element rotation vector  $r_c \in \mathbb{R}^3$  ( $\theta_c = \|r_c\|$ , axis  $r_c/\|r_c\|$ ). The 12 additional parameters  $(r_L, t_L, r_R, t_R)$  bring the total to 26.

Jointly fitting the 14 telecentric parameters and the 12 SE(3) parameters against the Zernike rayfield reduces pixel RMS from 14.6 px to 1.06 px (P50=0.87 px, P95=1.84 px)—a  $14\times$  improvement. The fitted rotations are  $\sim 2.5^\circ$  (left) and  $\sim 3.7^\circ$  (right); translations are sub-mm (Figure 6). Effective arm

angles of this magnitude in a precision CMO instrument are best read not as a macroscopic mechanical flexure of the microscope head but as the cumulative assembly tolerances of the internal relay optics (prisms, field lenses) between the shared objective and the two sub-pupils, whose small decentrations and tilts the SE(3) term absorbs as an equivalent rigid rotation of each ray bundle. The rotation angles are stable across optimisation runs; the translation components trade off with the telecentric base parameters (a known identifiability limitation).



**Figure 6.** Per-channel SE(3) arm alignment (schematic). The ideal telecentric CMO skeleton (dashed) places two off-axis sub-pupils  $S_L, S_R$  on the baseline  $b$ , emitting chief rays that converge on the optical axis at the object point  $O$  at working distance  $WD$  with full convergence angle  $\Theta$ . The fitted rigid SE(3) rotation tilts each optical arm away from this ideal by  $\sim 2.5^\circ$  (left) and  $\sim 3.7^\circ$  (right); the tilt is drawn  $\times 4$  for visibility. The rotation is a rigid SE(3) of the whole ray bundle, dominated by its  $x$ -component (fitted per-axis values: left  $(-2.3, -0.7, 0.6)^\circ$ , right  $(-2.6, -2.7, 0.2)^\circ$ ). Such effective angles arise from assembly tolerances of the internal relay optics between the shared objective and the sub-pupils, not from a macroscopic flexure of the microscope head.

### 3.6 Model Selection

Ray-space BIC [27] identifies the correct optical family among five candidates (pinhole, Brown-Conrady, parallel-plate, telecentric CMO with/without shear). The telecentric CMO+shear model wins by  $> 40\,000$  BIC points over all alternatives.

However, this 14-parameter model produces 14.6 px reprojection—it is the correct *family* but not a *usable* model. We introduce an **operational usability score** that adds a hard reprojection guard:

$$S_{\text{usable}} = \text{BIC}_{\text{ray}} + \mathbf{1}_{e_{\text{px}} > 1.5 \text{ px}} \left[ 10^6 + N_{\text{px}} \log \left( \frac{e_{\text{px}}^2}{1.5^2} \right) \right]. \quad (9)$$

Applying this score to every candidate rejects the 14p telecentric model ( $S_{\text{usable}} \approx +978\,890$ ) and selects the 26p SE(3)-aligned model as the best usable physical model ( $S_{\text{usable}} \approx -32\,433$ ); the full usability-filtered comparison is reported with the results in §4.10 (Table 11 and Figure 11). The 1.5 px threshold is an engineering convenience, not a tuned hyperparameter: the usable model (1.06 px) and the rejected families ( $\geq 14.6$  px for the 14p telecentric,  $> 300$  px for OpenCV) are separated by more than an order of magnitude, so any guard in the range  $\sim 2$ –14 px yields the identical selection.

A gauge-regularisation sweep confirmed negligible  $Z_0$ -direction drift after double TPS preprocessing: even the unregularised full-pose fit already has negligible gauge drift ( $0.023^\circ$ ), so explicit regularisation is unnecessary.

### 3.7 Why Direct Bundle Adjustment is Ill-Conditioned for Non-Central Optics

The two-stage decomposition strategy (rayfield measurement  $\rightarrow$  model identification) is motivated by a fundamental conditioning problem in direct bundle adjustment (BA). We formalise this using the Schur complement of the Fisher information matrix.

#### 3.7.1 Conditioning of the Joint Estimation Problem

Direct BA jointly estimates optical parameters  $\theta \in \mathbb{R}^p$  and board poses  $\eta \in \mathbb{R}^q$  ( $q = 6P$  for  $P$  frames) by minimising the pixel reprojection error. The Fisher information matrix at the optimum has the block structure:

$$\mathcal{I}(\theta, \eta) = \begin{bmatrix} \mathcal{I}_{\theta\theta} & \mathcal{I}_{\theta\eta} \\ \mathcal{I}_{\eta\theta} & \mathcal{I}_{\eta\eta} \end{bmatrix} = J^T J, \quad (10)$$

where  $J = \partial r / \partial(\theta, \eta)$  is the residual Jacobian. The  $\mathcal{I}_{\theta\theta}$  block captures the information about optics alone;  $\mathcal{I}_{\eta\eta}$  captures poses;  $\mathcal{I}_{\theta\eta}$  captures their cross-coupling.

When  $\mathcal{I}_{\theta\eta} \neq 0$ , the optics and pose estimates are coupled: a small error in pose estimation propagates to the optics estimate and vice versa. The effective information about  $\theta$  after marginalising over the uncertain poses is the **Schur complement**:

$$\mathcal{I}_{\theta|\eta} = \mathcal{I}_{\theta\theta} - \mathcal{I}_{\theta\eta} \mathcal{I}_{\eta\eta}^{-1} \mathcal{I}_{\eta\theta}. \quad (11)$$

We define the **coupling norm**  $c \in [0, 1]$  as the fraction of the optics information that is lost to pose coupling:

$$c = \frac{\|\mathcal{I}_{\theta\eta} \mathcal{I}_{\eta\eta}^{-1} \mathcal{I}_{\eta\theta}\|_F}{\|\mathcal{I}_{\theta\theta}\|_F}, \quad (12)$$

where  $\|\cdot\|_F$  is the Frobenius norm. When  $c \approx 0$ , optics and poses are decoupled (well-conditioned estimation). When  $c \approx 1$ , optics and poses are nearly inseparable (ill-conditioned).

### 3.7.2 Empirical Coupling Values

The Zernike rayfield itself (57 parameters, 0.47 px RMS) serves as a generic non-parametric reference: it is a per-pixel spline in the Zernike basis and makes no assumptions about the optical architecture. At the rayfield-identification stage, the compact 26p model loses approximately 0.6 px relative to this ceiling. This 0.6 px gap represents, for this dataset and model family, the cost of physical interpretability: the 26 parameters capture the dominant telecentric skeleton and SE(3) arm misalignment, while the remaining 0.6 px come from low-amplitude aberrations (field curvature, astigmatism, detection noise) that only the flexible 57-parameter Zernike field can absorb. For applications where sub-pixel accuracy is less critical than having a physically meaningful model, this trade-off is acceptable.

In the StereoComplex direct-BA diagnostic implementation, the real Pycaso configuration yields a Schur-complement coupling norm  $c = 0.98$ , while a synthetic CMO-like oracle gives  $c \approx 0.81$ , both computed from finite-difference Jacobians (forward differences, step  $\delta = 10^{-6}$ , parameters scaled to comparable units). Because such block norms depend on the parameter scaling, the values are reported as conditioning diagnostics rather than as universal dimensionless properties. The qualitative interpretation is robust:  $c$  near unity indicates strong optics–pose coupling, not a universal property of all CMO microscopes. That the real instrument ( $c = 0.98$ ) exceeds the idealised synthetic oracle ( $c \approx 0.81$ ) is consistent with the real CMO carrying optical aberrations and mechanical misalignments absent from the synthetic model, which add to the optics–pose coupling; only the qualitative “near unity” conclusion—not the precise value—should be read into either number. A sensitivity analysis varying each parameter group’s scale by  $\pm 1$  order of magnitude confirms that  $c$  is invariant under uniform rescaling of all parameters and varies by at most  $\Delta c = 0.26$  under per-group perturbations (worst case: shear coefficients); across the full sweep  $c$  remains above 0.72, well within the strong-coupling regime. This means that 81–98 % of the apparent information about optical parameters is absorbed by pose coupling (depending on the diagnostic configuration)—the two parameter blocks are nearly inseparable. By contrast, in the rayfield-mediated pipeline the coupling norm  $c = 0.0$  applies specifically to the pose-free ray-space model identification stage: the Zernike BA still estimates a constrained set of board poses (shared-rotation model, 15 pose parameters), but the subsequent physical model identification consumes the rayfield as a fixed observable and estimates no poses.

Figure 7 shows the normalised eigenvalue spectrum of the Schur complement  $S_\theta$  on the 26-parameter CMO optical block, evaluated at the rayfield-initialised Pycaso solution. Of the 26 eigenvalues, the trailing modes (red crosses,  $\lambda_i/\lambda_{\max} < 10^{-3}$ ) collapse after pose marginalisation—these are precisely the directions penalised by the Schur prior (Section 3.9). The coupling norm for the real Pycaso configuration is  $c = 0.98$  (synthetic CMO-like oracles give  $c \approx 0.81$ ), confirming that 98 % of the apparent optical information is absorbed by pose coupling. Schur-coupling values for the other oracles (pinhole, Brown-Conrady, parallel-plate, Greenough) were not reliably computable from the current diagnostic infrastructure and are therefore not reported.

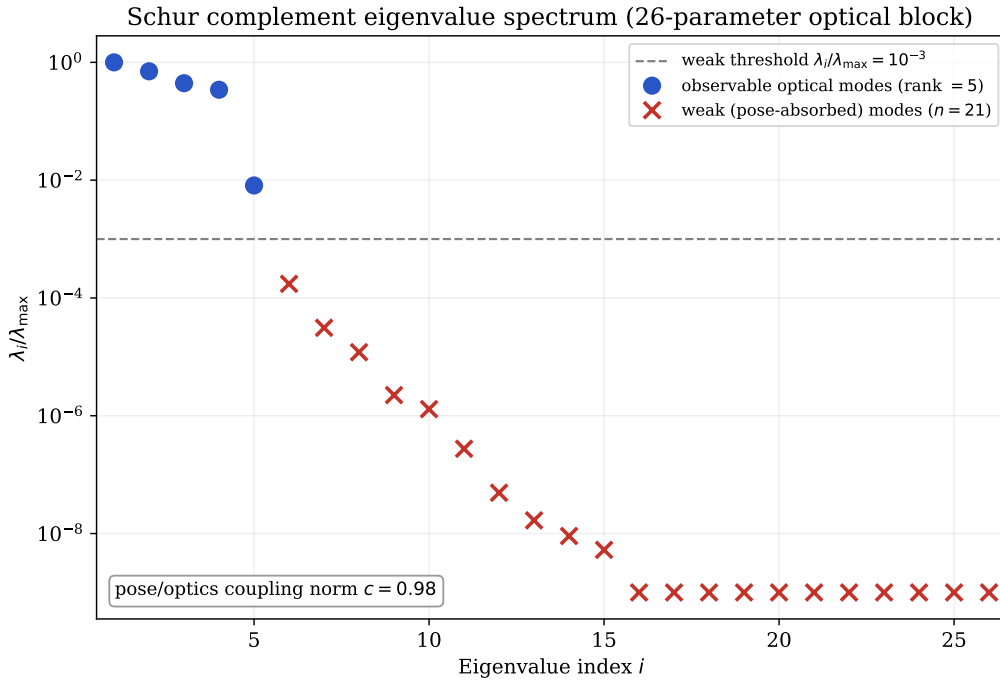
To assess which individual parameters are identifiable after pose marginalisation, we project each of the 26 unit parameter directions onto the strong subspace (the span of eigenvectors with  $\lambda_i/\lambda_{\max} \geq 10^{-3}$ ). The fraction of variance retained defines an **identifiability score**  $\in [0, 1]$ . Parameters with score  $> 0.5$  are considered *identified* (well-observed independently of poses); parameters with score  $< 0.1$  are *effective* (their value is driven by the gauge and regularisation, not by the data).

Only four parameters are identified: the shear coefficients  $s_x^L, s_y^L, s_x^R, s_y^R$  (scores 0.89–0.92). These



encode the dominant telecentric asymmetry and are robustly determined. Two SE(3) x-rotations ( $rv_L^x$ ,  $rv_R^x$ , scores  $\sim 0.28$ ) and the two y-translations ( $t_L^y$ ,  $t_R^y$ , scores  $\sim 0.22$ ) are partially identifiable. All other parameters—including the effective axial length  $f_{\text{obj}}$ , the angular focal length  $f_{\text{angular}}$ , the working distance  $WD$ , the baseline  $b$ , the principal point  $(c_x, c_y)$ , and the remaining SE(3) components—fall below 0.1: their individual values are effective rather than absolute, and only the linear combinations captured by the strong modes carry information independent of the pose gauge.

This confirms that the 26-parameter model provides *effective relative identification*: the compact physical skeleton (shear, telecentric structure) is data-supported, while the geometric scale ( $WD$ ,  $b$ ,  $f_{\text{obj}}$ ) and most SE(3) components are absorbed by the rayfield gauge and pose coupling. The term “identification” in this work refers to the ray-space residual structure that selects the CMO family and its dominant asymmetries, not to the absolute metrological determination of every individual parameter.



**Figure 7.** Normalised eigenvalue spectrum of the Schur complement  $S_\theta$  on the 26-parameter CMO optical block, computed from the rayfield-initialised Pycaso model. Eigenvalues below the weak threshold  $\lambda_i/\lambda_{\max} < 10^{-3}$  (red crosses) are the poorly observable directions after pose marginalisation—these are the modes penalised by the Schur prior in Section 3.9. The coupling norm for this configuration is  $c = 0.98$  (synthetic CMO-like oracles give  $c \approx 0.81$ ; see Section 3.7).

### 3.8 Corner Bundle Adjustment

The 26p model, fitted to the rayfield, is refined by direct corner bundle adjustment (joint optimisation of model parameters and per-frame poses on 3300 corner observations, 200 nfev, soft-L1 loss). The optimisation is performed in ray space; pixel RMS is computed afterwards by numerical inverse projection (Appendix A). The pixel RMS improves from 1.06 px to 0.88 px (P50=0.64 px, P95=1.70 px; 17%), while all 26 model parameters remain unchanged to within  $10^{-3}$  of their rayfield-fit values. The model geometry—effective baseline, working distance, effective axial length, and convergence angle (all in the rayfield gauge and fixed  $f_x$  reference), as well as the SE(3) arm corrections—is essentially identical before and after corner refinement. This confirms that the rayfield-initialised parameters are already



near-optimal for corner reprojection: the corner BA improves pose estimates but does not alter the physical model. This is a strong validation of the two-stage decomposition strategy: the rayfield fit, which never directly minimises corner error, produces parameters so close to the constrained-pose corner optimum that the dedicated constrained-pose BA can barely improve them, while also providing the correct basin and observability prior for the free-pose Schur-stabilised refinement. In contrast, the direct approach—jointly estimating optics and poses from corners—is severely ill-conditioned for this non-central system (Schur-complement coupling norm  $c = 0.98$  on the real Pycaso configuration,  $c \approx 0.81$  on synthetic CMO-like oracles), failing to converge with a pinhole initialisation (RMS 29 px).

### 3.9 Stabilising Direct BA with a Schur-Complement Prior

The corner BA of Section 3.8 improves the 2D pixel reprojection RMS by 17 % without altering the physical model. However, the Schur-complement diagnostic of Section 3.7 reveals that the Fisher information matrix at the rayfield solution  $\theta_0$  has weak eigenmodes—directions in parameter space where a small change in optics can be nearly perfectly compensated by a change in board poses. An unregularised BA can drift along these modes, trading physical interpretability for marginal reprojection gains.

#### 3.9.1 Schur-Eigenmode Prior

The Schur complement at the rayfield solution is diagonalised,

$$S_0 = S_\theta(\theta_0, \eta_0) = V\Lambda V^T, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_{n_\theta}), \quad \lambda_1 \geq \dots \geq \lambda_{n_\theta}, \quad (13)$$

where  $V = [v_1, \dots, v_{n_\theta}]$  are the eigenvectors and  $\lambda_i$  the eigenvalues. Directions  $v_i$  with small  $\lambda_i/\lambda_{\max}$  are poorly observable after pose marginalisation.

The prior penalises displacement along each eigenmode with a weight inversely proportional to its eigenvalue [32]:

$$\mathcal{L}_{\text{Schur}}(\theta) = \alpha \sum_{i=1}^{n_\theta} w_i \left( v_i^T D_\theta^{-1} (\theta - \theta_0) \right)^2, \quad w_i = \left( \frac{\lambda_{\max}}{\lambda_i + \varepsilon \lambda_{\max}} \right)^p, \quad (14)$$

where  $D_\theta = \text{diag}(\sigma_1, \dots, \sigma_{n_\theta})$  contains per-parameter scales (degrees for rotations, millimetres for translations, pixels for the principal point),  $\alpha$  controls the overall prior strength,  $p$  the sharpness of the eigenvalue weighting ( $p = 1$  throughout), and  $\varepsilon = 10^{-6}$  prevents division by zero. The prior is added to the least-squares objective as  $n_\theta$  extra residual rows  $r_i^{\text{prior}} = \sqrt{\alpha w_i} v_i^T D_\theta^{-1} (\theta - \theta_0)$ .

#### 3.9.2 Prior Sweep and Comparison with Tikhonov Baseline

**Two BA configurations must be distinguished.** Section 3.8 reports a *constrained-pose* corner BA: the board rotation is shared across frames and only the Z translation varies, yielding  $1.06 \text{ px} \rightarrow 0.88 \text{ px}$  (Table 10) with 15 pose parameters. The sweep in this section intentionally *releases the pose constraint*: each of the 10 frames has a free 6-DOF pose (60 pose parameters), exposing the full optics–pose coupling identified by the Schur diagnostic of Section 3.7. The resulting unregularised RMS (0.241 px) is lower than the constrained-pose BA precisely because the free poses can absorb residual rayfield errors—which is exactly the compensation the Schur prior is designed to prevent. The two RMS values should not be conflated: the 0.88 px is the operational calibration result; the 0.241 px is a diagnostic upper bound

obtained by deliberately exposing the ill-conditioned coupling.

An isotropic (Tikhonov) prior  $\mathcal{L}_{\text{iso}} = \alpha \|D_{\theta}^{-1}(\theta - \theta_0)\|^2$  serves as a baseline. Both priors are swept over  $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$  with free per-frame poses (200 nfev, soft-L1 loss). The 2D reprojection RMS is computed by numerical projection (Appendix A).

**Table 2.** Regularisation sweep: isotropic (Tikhonov) vs. Schur-complement prior.  $\Delta_{\text{weak}}$  is the norm of the scaled parameter displacement projected onto the weak Schur eigenspace ( $\lambda_i/\lambda_{\text{max}} < 10^{-3}$ ).  $\checkmark/\times$  indicates convergence.

| Isotropic prior |                        |                        | Schur prior |                        |                        |
|-----------------|------------------------|------------------------|-------------|------------------------|------------------------|
| $\alpha$        | RMS (px)               | $\Delta_{\text{weak}}$ | $\alpha$    | RMS (px)               | $\Delta_{\text{weak}}$ |
| 0               | 0.241 ( $\checkmark$ ) | 0.934                  | 0           | 0.241 ( $\checkmark$ ) | 0.934                  |
| $10^{-4}$       | 0.239 ( $\times$ )     | 0.529                  | $10^{-4}$   | 0.277 ( $\checkmark$ ) | <b>0.0033</b>          |
| $10^{-3}$       | 0.245                  | 0.248                  | $10^{-3}$   | 0.278                  | <b>0.0007</b>          |
| $10^{-2}$       | 0.257                  | 0.082                  | $10^{-2}$   | 0.279                  | <b>0.0001</b>          |
| $10^{-1}$       | 0.270                  | 0.017                  | $10^{-1}$   | 0.279                  | $< 10^{-4}$            |
| $10^0$          | 0.278                  | 0.003                  | $10^0$      | 0.283                  | $< 10^{-4}$            |
| $10^1$          | 0.286                  | 0.010                  | $10^1$      | 0.323                  | $< 10^{-4}$            |

Table 2 reports the results. The  $\alpha=0$  row is the unregularised direct BA: it converges at 0.241 px but at the cost of  $\Delta_{\text{weak}} = 0.934$ , i.e. essentially *all* of the parameter drift lives in the weak Schur subspace—the optimiser exploits the poorly observable directions to chase the data residual. The isotropic prior faces a classical trade-off when trying to suppress that drift: a small  $\alpha$  still leaves the weak modes loose ( $\Delta_{\text{weak}} = 0.53$  at  $\alpha=10^{-4}$ , where the optimiser also fails to converge), while a large  $\alpha$  degrades the fit (RMS rises to 0.286 px). The Schur prior breaks this trade-off: even at  $\alpha=10^{-4}$  it suppresses 99.6 % of the weak-mode drift ( $0.934 \rightarrow 0.0033$ , a  $280\times$  reduction) while keeping the RMS within 0.038 px of the unregularised baseline. At  $\alpha=10^{-3}$  the weak-mode drift is below  $10^{-3}$  and the RMS penalty is only 0.038 px—the prior blocks the poorly observable directions while leaving the well-observed degrees of freedom free to improve the fit. At the strongest setting ( $\alpha=10$ ), the Schur prior still converges, while the isotropic prior produces a higher RMS (0.286 px vs 0.323 px) with residual weak drift (0.010).

### 3.9.3 The Double Role of the Rayfield

This result formalises a double role for the rayfield estimate  $\theta_0$  in direct bundle adjustment:

1. **Initialiser.**  $\theta_0$  places the direct BA in the correct convergence basin (Section 3.8), avoiding the local minima that trap a pinhole or perspective-CMO initialisation (RMS 29 px vs. 0.88 px).
2. **Observability prior.** The Schur eigenmodes of the Fisher matrix at  $\theta_0$  tell the optimiser *which directions it may trust*. The prior blocks compensation between poses and intrinsics without penalising the genuinely observable optical degrees of freedom.

The Schur prior is not a generic regulariser—it is *derived from the same data* that produced the initialisation. The rayfield’s Fisher information matrix, evaluated at  $\theta_0$ , encodes the local observability structure of the inverse problem. Using that structure to guide the subsequent BA is the methodological contribution: the rayfield is not merely a calibration model but an *observability oracle* for direct refinement.

## 4 Results

### 4.1 Data Sources

Table 3 clarifies which results come from real Pycaso calibration data, which from synthetic oracles, and which are methodological diagnostics.

**Table 3.** Data sources for results reported in this paper. Real data: 10 Pycaso stereo pairs. Synthetic oracles: StereoComplex model\_selection\_oracles.py. Method: analytical and numerical diagnostics independent of data source.

| Result category          | Source                              | Examples                                                                                                                                      |
|--------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Real data</b>         | Pycaso CMO images (10 stereo pairs) | Zernike fit (0.47 px), CMO+SE(3) (1.06 px), descriptors ( $b$ , $WD$ , $\theta$ ), pose sweep, cross-validation, bootstrap, $f_x$ sensitivity |
| <b>Synthetic oracles</b> | StereoComplex simulator (noiseless) | Direct vs. rayfield inversion ( $c = 0.81$ ), ray-space BIC, oracle parameter recovery                                                        |
| <b>Methodological</b>    | Analytical / numerical diagnostics  | BIC formula, Plücker residual decomposition, SE(3) ablation, Schur derivation                                                                 |

### 4.2 Dataset and Pipeline

Ten stereo pairs of  $2048 \times 2048$  px ChArUco [9, 22] images ( $16 \times 12$  squares, 0.3 mm, DICT\_6X6\_250, setLegacyPattern(True)) from the Pycaso open dataset [8] are used. The Z-range is 2.65 mm–3.35 mm ( $\Delta = 0.70$  mm). Corner detection yields 61–161 corners per frame; Hessian-based completion ( $|\det H| + \text{Otsu} + \text{barycentre}$ ) fills all 165. A double Ray2D TPS denoising pass [3] (ArUco markers at  $\lambda = 10$ , then completed 165 corners at  $\lambda = 3$ , Huber  $c = 1.5$ ) produces gauge-stable 2D corners. The two-phase TPS is not cosmetic: without it, raw ChArUco detections carry structured biases from compression artifacts, blur, and defocus that cause the Zernike rayfield fit to become gauge-unstable—the direction piston  $Z_0$  drifts by  $8.5^\circ$  between constrained and full-pose fits, making physical interpretation impossible. After double TPS, the gauge drift drops to  $0.023^\circ$ , and the rayfield becomes a stable experimental oracle [33].

### 4.3 Pose Count Sensitivity

To assess robustness to the number of calibration frames, we sub-sample  $N \in \{3, 5, 8, 10\}$  frames from the 10 available and re-run the full pipeline (Zernike fit + 26p model) for each. Table 4 reports pixel RMS and effective geometric descriptors.

**Table 4.** Pose sweep: pixel RMS and effective-descriptor stability vs. number of frames (all descriptors in the rayfield gauge with  $f_x = 25600$  px). Each row is an independent refit on  $N$  frames; the  $N=10$  value (1.07 px) reproduces the main-text calibration (1.06 px, full pipeline) to within refit noise.

| $N$ frames | RMS (px) | P50 (px) | $b$ (mm) | $WD$ (mm) | $\theta$ ( $^\circ$ ) |
|------------|----------|----------|----------|-----------|-----------------------|
| 3          | 1.02     | 0.84     | 24.7     | 64.6      | 22.4                  |
| 5          | 1.03     | 0.84     | 24.8     | 64.6      | 22.5                  |
| 8          | 1.06     | 0.87     | 24.9     | 64.7      | 22.6                  |
| 10         | 1.07     | 0.88     | 24.8     | 64.7      | 22.4                  |

The pixel RMS is stable within 0.05 px across frame counts—even 3 frames suffice for 1.02 px

calibration. Effective descriptors vary by less than 0.2 mm (baseline) and 0.1 mm (working distance) in the chosen gauge, confirming that the pipeline is robust to the number of calibration images. The effective convergence angle  $\theta \approx 22.5^\circ$  (standard deviation  $0.1^\circ$ ).

#### 4.4 OpenCV Tuning Does Not Recover

We tested the full OpenCV stereo calibration flag space: `CALIB_RATIONAL_MODEL` (8 distortion coefficients), `CALIB_THIN_PRISM`, and `CALIB_TILTED_MODEL`. None of these configurations produced a usable calibration on the CMO data. The rational model, with the most parameters (14 intrinsics per camera), still fails because the distortion model remains radial and centred—it cannot represent a non-central projection where rays originate from two different sub-pupils. OpenCV’s fisheye model (Kannala–Brandt [12]) was not pursued for the same structural reason: despite its strong radial-distortion capacity it is still a *central* projection with a single optical centre, and so cannot represent the two-sub-pupil geometry either. The best OpenCV configuration achieves  $>300$  px RMS, compared to 1.06 px for our 26-parameter CMO+SE(3) model. This confirms that the failure is structural (central projection assumption), not a matter of under-parameterised distortion.

#### 4.5 Geometric Descriptor Consistency

The manufacturer documentation identifies the instrument as a Leica M205C common-main-objective (CMO) stereomicroscope, equipped with a Planapo  $1.0\times$  objective, an HD-V photo/video tube, a  $1.0\times$  video/photo objective and a  $1.0\times$  C-mount adapter. For the Planapo  $1.0\times$  configuration, Leica specifies a working distance of 61.5 mm [13]. The datasheet does *not*, however, disclose the effective CMO stereo baseline, the sub-pupil positions, or the convergence angle—the internal geometry of the shared-objective design. These quantities are recovered here from the Zernike rayfield rather than read from the documentation, which is precisely where the rayfield is informative.

A quantitative consistency check follows from the chief-ray geometry of the telecentric CMO model used in this work. The two stereo channels view the object through two off-axis sub-pupils separated by a baseline  $b$  in the exit-pupil plane of the shared objective. By the focal (Fourier) property of the objective, a pupil point at transverse height  $y$  maps to a chief ray that leaves object space at an angle  $\theta_y$  with  $\tan \theta_y = y/f_{\text{obj}}$ , where  $f_{\text{obj}}$  is the effective axial length (the sub-pupil-plane to working-plane distance in the CMO chief-ray model). The two sub-pupils at  $\pm b/2$  therefore emit chief rays inclined at  $\pm \arctan(b/(2f_{\text{obj}}))$  to the optical axis, so the full convergence angle obeys

$$\Theta = 2 \arctan \frac{b}{2f_{\text{obj}}}, \quad \text{equivalently} \quad b = 2 f_{\text{obj}} \tan \frac{\Theta}{2}, \quad (15)$$

in which the relevant axial length is the sub-pupil-plane-to-working-plane distance  $f_{\text{obj}} = WD - z_p$  in the CMO chief-ray model; we do not require it to be interpreted as the manufacturer objective focal length.

The convergence angle  $\Theta = \arccos(d_L \cdot d_R) = 22.6^\circ$  is a *scale-invariant* rayfield observable—it depends only on the reconstructed ray directions, not on the chosen reference scale. Together with the reconstructed  $b = 24.9$  mm,  $WD = 64.7$  mm and  $z_p = 2.5$  mm (hence  $f_{\text{obj}} = 62.2$  mm), the descriptors satisfy (15) internally:  $2 \times 62.2 \text{ mm} \times \tan(11.3^\circ) = 24.9 \text{ mm} = b$ , confirming that the centre-pixel rayfield obeys the CMO chief-ray geometry. The *external* check uses the only absolute length the datasheet provides. Adopting the documented working distance as the axial scale,  $f_{\text{obj}} \approx WD_{\text{doc}} = 61.5$  mm, and

the measured  $\Theta$ , relation (15) predicts

$$b_{\text{pred}} = 2 \times 61.5 \text{ mm} \times \tan(11.3^\circ) = 24.6 \text{ mm}, \quad (16)$$

within 0.3 mm ( $\approx 1\%$ ) of the reconstructed  $b = 24.9$  mm; equivalently, the reconstructed baseline and the documented working distance imply  $\Theta_{\text{pred}} = 2 \arctan(24.9/(2 \times 61.5)) = 22.9^\circ$ , against the measured  $22.6^\circ$ . The blind, Zernike-informed reconstruction thus recovers a CMO geometry quantitatively compatible with the documented Planapo 1.0 $\times$  configuration.

This agreement is a strong physical *consistency* check, not an independent metrological validation of the Leica specifications. The convergence angle is scale-invariant, but the absolute lengths ( $b$ ,  $WD$ ,  $f_{\text{obj}}$ ) inherit the chosen reference focal length  $f_x = 25\,600$  px, and  $WD$  scales linearly with it (§4.7); reproducing the documented  $WD$  to a few percent shows only that this scale choice is compatible with the documented optics, since it still relies on the datasheet  $WD$  as the scale anchor rather than on an independent 3D measurement. What the test does establish is that the rayfield recovers internal CMO parameters absent from the manufacturer documentation, and that they are quantitatively consistent with the documented Planapo 1.0 $\times$  configuration. Beyond this external check, the descriptors are also **internally consistent** by three independent tests: (1) cross-validation stability:  $1.07 \pm 0.02$  px RMS across 5 folds, with effective-descriptor spread  $\pm 0.4$  mm (baseline),  $\pm 0.15$  mm ( $WD$ ); (2) bootstrap 95 % confidence intervals:  $b \in [23.48, 25.04]$  mm,  $WD \in [64.64, 65.13]$  mm,  $\theta \in [21.04^\circ, 22.72^\circ]$ ; (3) frame-count stability: effective descriptors vary by  $< 0.2$  mm (baseline) and  $< 0.1$  mm ( $WD$ ) from 3 to 10 frames (Table 4).

An independent *absolute* validation—a certified 3D artefact at a known distance, or a direct measurement of the baseline or working distance—remains open and is listed among the limitations (Section 5.2). The present result nonetheless demonstrates the tool’s value: from images alone, the rayfield recovers the undisclosed internal CMO geometry and places it in quantitative agreement with the manufacturer’s documented Planapo 1.0 $\times$  configuration.

#### 4.6 Comparison with the Soloff Polynomial Baseline

To calibrate the accuracy–interpretability trade-off, we benchmark against the Soloff polynomial model [28], a standard approach in stereo-microscope calibration [8]. A Soloff model maps 3D world coordinates  $(X, Y, Z)$  directly to pixel coordinates  $(u, v)$  via a polynomial expansion  $u = \sum a_i \phi_i(X, Y, Z)$ ,  $v = \sum b_i \phi_i(X, Y, Z)$  using monomials  $\phi_i$  up to a chosen degree. Four independent polynomials are fitted (left- $u$ , left- $v$ , right- $u$ , right- $v$ ) by ridge-regularised least squares on the same ChArUco corner observations used for our Zernike rayfield—identical input data, identical train/evaluation split. The model has no notion of 3D rays, no geometric consistency constraint between channels, and no optical interpretation of its coefficients.

Table 5 reports the pixel reprojection RMS for Soloff fits of degree 1 to 4 alongside the Zernike rayfield and the CMO 26p model.

**Table 5.** Pixel reprojection RMS on the Pycaso dataset: Soloff polynomial vs. Zernike rayfield vs. CMO 26p. All models are evaluated on the same 1,650 stereo corner correspondences (10 frames  $\times$  165 corners), i.e. 3,300 image observations over the two channels. Soloff fits are ridge-regularised ( $\lambda = 10^{-3}$ ). The Zernike RMS floor ( $\sim 0.41$  px) is the best achievable under the 3D geometric consistency constraint; the Soloff floor ( $\sim 0.18$  px) is the best achievable by unconstrained 2D regression. The 0.23 px gap quantifies the cost of enforcing a physically meaningful 3D ray model.

| Model               | Params | RMS (px) | Interpretable |
|---------------------|--------|----------|---------------|
| Soloff deg. 1       | 16     | 1.66 px  | No            |
| Soloff deg. 2       | 40     | 0.22 px  | No            |
| Soloff deg. 3       | 80     | 0.18 px  | No            |
| Soloff deg. 4       | 140    | 0.18 px  | No            |
| Zernike O(0)+d(2)   | 57     | 0.47 px  | Rayfield      |
| Zernike O(1)+d(3)   | 93     | 0.41 px  | Rayfield      |
| CMO telecentric 14p | 14     | 14.6 px  | Yes           |
| CMO 26p (SE(3))     | 26     | 1.06 px  | Yes           |

Three observations follow. First, the Soloff polynomial is a more accurate pixel regressor than any 3D-geometric model: at comparable parameter counts, Soloff deg. 2 (40 params, 0.22 px) outperforms Zernike O(0)+d(2) (57 params, 0.47 px) by a factor of  $\sim 2.1\times$ , and Soloff deg. 3 (80 params, 0.18 px) outperforms Zernike O(1)+d(3) (93 params, 0.41 px) by  $\sim 2.3\times$ . The polynomial reaches a noise floor of 0.18 px at degree 3, beyond which additional parameters ( $\geq 140$ ) produce no further improvement.

Second, the Zernike rayfield reaches its own floor at 0.41 px, approximately 0.23 px above the Soloff floor. This gap quantifies, for this dataset and model family, the cost of enforcing 3D geometric consistency: the rayfield must produce a per-pixel 3D line whose intersections with rays from the other channel are geometrically meaningful, whereas Soloff merely interpolates 2D coordinates. The gap is not a failure of the Zernike basis—it is a physical consequence of solving a harder problem.

Third, the CMO 26p model loses an additional 0.65 px relative to the Zernike reference, but gains a physically interpretable parameterisation: its 26 parameters are the baseline, working distance, effective axial length, convergence angle, shear coefficients, and per-arm SE(3) poses. No Soloff coefficient can be read as a baseline or a working distance.

These figures are the model *floors at construction*, not the endpoint. The 1.06 px of the CMO 26p model is its rayfield-identified *initialisation*; once that initialisation seeds the free-pose Schur-stabilised direct bundle adjustment (§3.9), the same 26 parameters refine to 0.28 px, narrowing the gap to the Soloff degree-3 floor to under 0.1 px. After refinement, then, the compact physical model is competitive in pixel accuracy with the black-box regressor while retaining a physically interpretable parameterisation—and it does so without the optics–pose compensation that the unregularised free-pose optimum (0.24 px) exploits.

The comparison thus frames the paper’s central trade-off explicitly. At construction: 0.18 px (black-box, no geometry), 0.41 px (geometric rayfield, no optical model), 1.06 px (compact physical model, physically interpretable parameterisation). After free-pose Schur-stabilised refinement the physical model closes most of that gap, reaching 0.28 px—so interpretability ultimately costs a fraction of a pixel rather than a category of accuracy.

This trade-off has a computational counterpart. A classical Soloff calibration is a linear least-squares fit of four polynomials and is essentially instantaneous (a few milliseconds on this dataset). The rayfield-to-



CMO identification is, by contrast, an iterative non-linear optimisation: jointly fitting the Zernike rayfield with the board poses and then the 26-parameter CMO model takes  $\approx 20$  s on a single CPU core for the ten-pose dataset, and the optional free-pose Schur-stabilised refinement adds a comparable amount. This is a one-time, offline calibration cost—three to four orders of magnitude above a Soloff fit, but negligible in absolute terms—and once calibrated both models evaluate a pixel in closed form. The pipeline thus spends a modest offline compute budget to obtain the physical descriptors and residual diagnostics that a polynomial cannot provide.

Beyond the accuracy–interpretability trade-off, the field-resolved rayfield does what a diagnostic instrument should: it points to the next physical degree of freedom missing from the compact model. The Zernike rayfield reveals that the real instrument is not perfectly telecentric: the effective sub-pupil positions shift laterally by approximately 1.0 mm across the field, producing a centre-to-edge baseline variation that the 26-parameter telecentric skeleton misses entirely (it holds the sub-pupils at fixed lateral positions). This residual non-telecentricity is a natural extension of the model; but the per-parameter identifiability analysis of §3.7 shows that any parameter controlling axial or lateral sub-pupil scale lies in the null space of the planar ChArUco observations and cannot be identified without an external reference. This is consistent with the specimen comparison (§4.8), where the residual CMO–Zernike difference reduces to a single depth-only scale factor ( $s_z \approx 0.70$ ) that planar data alone cannot resolve. The rayfield thus fulfills its diagnostic role not by claiming completeness, but by identifying exactly which degree of freedom remains unresolved—and under what conditions it could be anchored.

#### 4.6.1 Extrapolation to Held-Out Poses

The accuracy–interpretability trade-off above is measured on the full 10-frame dataset (train = test, since both models use all available data). A stricter question is: how well does each model *extrapolate* to poses not seen during fitting? This tests whether the 3D geometric constraint in the Zernike rayfield provides a regularisation benefit beyond what the Soloff polynomial’s ridge penalty achieves.

We split the 10 frames into 7 training poses (the most central ones, closest to the centroid of the pose distribution) and 3 held-out test poses (the most extreme, at the edges of the sampled volume). Both models are fitted on the training set only and evaluated on the held-out frames. Critically, the two model families have different natural output spaces—Soloff maps 3D coordinates to pixels, Zernike maps pixels to 3D rays—so their extrapolation errors are measured in their respective native metrics and compared via the *degradation ratio* (test error / train error).

For Soloff, the native metric is the 2D pixel prediction error (the RMS distance between the polynomial-predicted pixel and the observed corner). For the Zernike rayfield, the native metric is the 3D triangulation accuracy: observed left and right pixels are converted to rays via the rayfield, the rays are intersected to obtain a 3D point, and this point is compared to the true corner position (known from the board model and the frame pose estimated by a Kabsch alignment of the triangulated points to the board). The Kabsch step removes the rigid gauge freedom inherent to uncalibrated stereo reconstruction (the absolute pose of the board cannot be recovered without an external reference), so the residual measures only the non-rigid rayfield quality.

Table 6 reports the results. Both models degrade by a factor of approximately  $1.9\times$  on held-out extreme poses, confirming that the 7-frame training set is sufficient to build a usable model in both cases. The crucial difference appears at higher model complexity: Soloff degree 3 (80 parameters) degrades



by a factor of  $4.65\times$  (from 0.16 px to 0.75 px): its extra high-order terms lower the in-sample residual but capture training-pose-specific structure that does not transfer to the held-out extreme poses—the operational signature of overfitting. We do not claim a single mechanism; the absorbed structure may be detection noise, pose-dependent systematic residuals, or high-order instability when extrapolating beyond the training pose distribution. The Zernike rayfield (180 parameters,  $O(4)+d(4)$ ), by contrast, degrades by only  $1.86\times$  despite having more than twice as many parameters. The 3D geometric consistency constraint—every pixel’s ray must intersect consistently with rays from the other channel and with the board geometry—acts as a structural regulariser that prevents the rayfield from memorising individual training poses.

**Table 6.** Extrapolation to held-out extreme poses (7 train / 3 test). Each model is evaluated in its native metric: 2D pixel error for Soloff (3D→2D mapping), 3D triangulation accuracy for Zernike (2D→3D mapping). The degradation ratio (test / train) is the comparable quantity across model families.

| Model               | Params | Train     | Test (extreme) | Degradation                    |
|---------------------|--------|-----------|----------------|--------------------------------|
| Soloff deg. 1       | 16     | 1.21 px   | 2.48 px        | $2.05\times$                   |
| Soloff deg. 2       | 40     | 0.20 px   | 0.39 px        | $1.97\times$                   |
| Soloff deg. 3       | 80     | 0.16 px   | 0.75 px        | <b><math>4.65\times</math></b> |
| Zernike $O(4)+d(4)$ | 180    | 0.0006 mm | 0.0012 mm      | <b><math>1.86\times</math></b> |

Two conclusions follow. First, at comparable degradation ratios ( $\sim 1.9\times$ ), Soloff remains more accurate in its native 2D pixel metric whereas the Zernike rayfield remains stable in its native 3D triangulation metric, consistent with the in-sample comparison of Table 5: the 2D polynomial is a more accurate regressor, but its accuracy collapses when the model order is pushed (degree 3). Second, the Zernike rayfield’s degradation stays bounded ( $1.86\times$ ) even at 180 parameters because the 3D geometric consistency constraint prevents it from overfitting to individual poses. The rayfield does not learn “pixel (500, 1300) corresponds to 3D point  $X$  in frame 3”; it learns “pixel (500, 1300) emits a ray that must intersect the corresponding ray from the other channel at a 3D point consistent with a rigid board”. This constraint is absent from the per-coordinate polynomial regression, which treats each pixel coordinate as an independent scalar function of  $(X, Y, Z)$ .

We deliberately do not convert the Zernike 3D error to a pixel-equivalent: that conversion factor depends on the local ray geometry (pixel pitch, working distance, ray obliquity) and the inverse map is, moreover, ill-conditioned at the microscope’s magnification ( $f_x = 25600$  px gives  $\partial d/\partial u \approx 2.5 \mu\text{m}/\text{px}$ , so a sub-millimetre ray miss displaces the inverted pixel by tens of pixels). The degradation ratio of Table 6 sidesteps this by comparing each family in its native metric.

A complementary and more direct comparison expresses *both* models in a single physical metric—the 3D reconstruction error in micrometres. This requires a shared board-pose reference: the absolute board pose is a gauge freedom that neither uncalibrated model fixes on its own, and comparing the Zernike reconstruction against the CMO-model poses alone would conflate the rayfield error with the sub-millimetre disagreement between the two bundle adjustments’ pose estimates (an unaligned comparison inflates the Zernike error to  $\sim 500 \mu\text{m}$ , almost entirely pose mismatch). We therefore freeze the poses and evaluate two symmetric protocols. (i) *CMO-pose reference*: the Zernike rayfield is refitted with the board poses held fixed at the CMO-model values  $\{R_i, t_i\}$ —only the 180 rayfield coefficients are optimised—so both models share the exact pose set that the Soloff polynomial implicitly uses; a single

global rigid (Kabsch) transform then removes the residual camera-to-world gauge of the triangulated points. (ii) *Zernike-pose reference*: conversely, the per-frame poses self-consistent with the free-fit Zernike field are recovered by pose-only least squares, and the four Soloff polynomials are refitted against that geometry. Reporting both protocols brackets the reference dependence and tests whether the conclusion survives the choice of which model defines the board geometry.

**Table 7.** 3D reconstruction RMS (micrometres) on held-out extreme poses, with both models expressed against a shared board-pose reference. Two symmetric protocols bracket the reference dependence: poses frozen to the CMO model, or to the free-fit Zernike field. A single global rigid (Kabsch) transform, estimated on the training frames, removes the camera-to-world gauge. The qualitative ranking is invariant to the reference choice.

| Model             | Params | CMO-pose ref.                |               | Zernike-pose ref.            |       |
|-------------------|--------|------------------------------|---------------|------------------------------|-------|
|                   |        | Train→Test ( $\mu\text{m}$ ) | Degr.         | Train→Test ( $\mu\text{m}$ ) | Degr. |
| Soloff deg. 2     | 40     | 1.24 → 2.14                  | 1.72×         | 2.58 → 4.78                  | 1.86× |
| Soloff deg. 3     | 80     | 0.99 → 3.81                  | <b>3.86</b> × | 1.99 → 3.87                  | 1.95× |
| Zernike O(4)+d(4) | 180    | 1.18 → 2.39                  | 2.04×         | 2.63 → 5.53                  | 2.10× |

Table 7 shows the ranking is invariant to the reference choice. In both protocols the 180-parameter Zernike rayfield *matches the accuracy of the 40-parameter Soloff degree-2 fit* (1.2  $\mu\text{m}$ /2.4  $\mu\text{m}$  train/test under the CMO reference, 2.6  $\mu\text{m}$ /5.5  $\mu\text{m}$  under the Zernike reference) and degrades by only  $\sim 2\times$ , never collapsing. Soloff degree 3 attains the lowest training error in both references, but its degradation is reference-dependent: against the cleaner CMO geometry (the CMO model fits all ten frames jointly) it overfits sharply (3.86 $\times$ ), consistent with its native pixel-metric collapse (4.65 $\times$ , Table 6); against the noisier Zernike-recovered geometry the same overfit is partly masked (1.95 $\times$ ) because the  $\sim 2\text{ }\mu\text{m}$  pose-recovery floor inflates the training residual. The absolute micrometre values and the degree-3 overfit magnitude therefore depend on which model’s poses define the reference—neither constitutes an external metrological truth, consistent with the absolute-scale caveats above—but the qualitative conclusion is robust across both: the Zernike rayfield buys 3D geometric consistency and overfit resistance at the cost of  $\sim 0.2\text{ px}$ , whereas a high-order Soloff polynomial maximises raw pixel accuracy at the risk of overfitting.

#### 4.7 Internal Validation

Three internal validation experiments assess the robustness of the 26p pipeline on the Pycaso dataset.

**Cross-validation.** 5-fold leave-2-out cross-validation (fit on 8 frames, evaluate on 2 held-out frames) yields a mean pixel RMS of  $1.07 \pm 0.02\text{ px}$ . The low standard deviation ( $\sim 0.02\text{ px}$ ) confirms that the model is not overfitting to specific frames; the calibration is stable across the 10-frame dataset. Fold 1 shows slightly higher RMS (1.09 px) and a lower effective baseline estimate ( $b = 23.1\text{ mm}$ ) because the held-out frames (0 and 1) cover the most extreme Z values.

**Bootstrap descriptor stability.** 100 bootstrap iterations (resampling 10 frames with replacement, refitting the full pipeline each time) give 95 % confidence intervals:

- Effective baseline  $b$ : [23.48, 25.04] mm (mean 24.56 mm,  $\sigma = 0.43\text{ mm}$ )
- Effective working distance  $WD$ : [64.64, 65.13] mm (mean 64.86 mm,  $\sigma = 0.15\text{ mm}$ )
- Effective convergence angle  $\theta$ : [21.04°, 22.72°] (mean 22.17°,  $\sigma = 0.46^\circ$ )

The effective working distance is the most stable descriptor ( $\sigma < 0.2$  mm), while the effective baseline and convergence angle show 0.4 mm and  $0.4^\circ$  variation respectively—consistent with the gauge sensitivity of the Zernike origin field noted in Section 5.2.

**Sensitivity to fixed focal length  $f_x$ .** Varying the fixed pinhole focal length  $f_x$  by  $\pm 10\%$  (5 values: 23040, 24320, 25600, 26880, 28160 px) produces:

- Pixel RMS:  $1.21 \rightarrow 0.95$  px—remains near the one-pixel range, confirming that the rayfield is stable as a calibration object.
- Working distance:  $58.3 \rightarrow 71.3$  mm—linearly proportional to the initial  $f_x$  ( $\pm 10\% f_x \rightarrow \pm 10\%$  WD). WD is directly scaled by the pinhole reference and is *not* independently identifiable.
- Convergence angle:  $24.1^\circ \rightarrow 20.5^\circ$ —varies inversely with WD.
- Baseline:  $23.9 \rightarrow 25.1$  mm ( $\pm 2.5\%$ )—moderately stable.

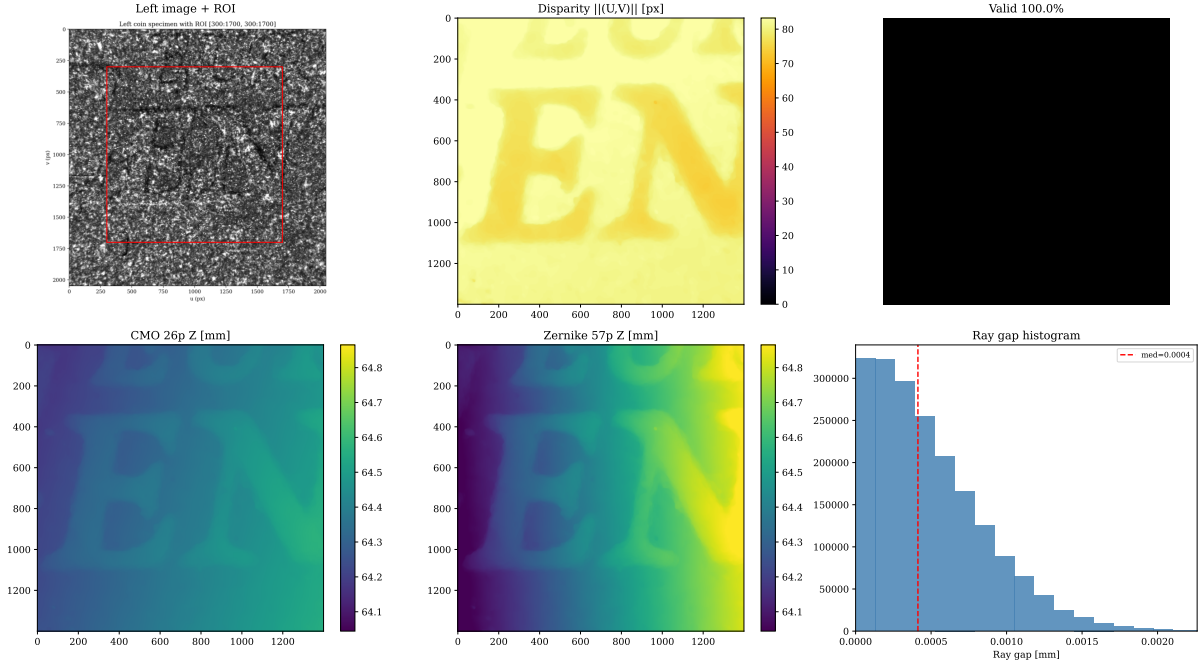
The absolute physical descriptors, especially  $WD$  and  $\theta$ , remain sensitive to the chosen pinhole reference scale. Descriptor readout therefore requires either an independently known optical scale (pixel pitch, magnification) or external metrological validation—a known limitation of all single-view calibration methods without a known reference object.

#### 4.8 Application Sanity Check: Specimen Reconstruction

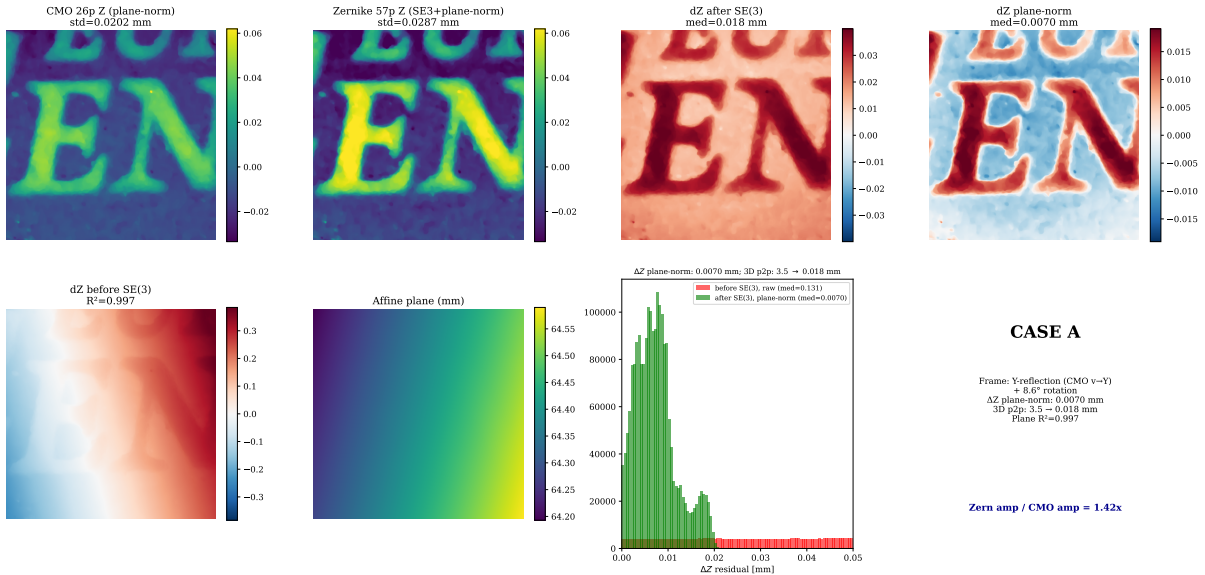
A non-metrological sanity check validates the calibrated rayfields on an independent speckled coin specimen imaged with the same Pycaso microscope (single stereo pair, no ChArUco pattern). This is **not** absolute metrological certification against a traceable 3D artefact, but—together with the external profilometry of the same specimen (§4.9)—it confirms that the rayfields produce coherent dense reconstruction on data unseen during calibration. Dense left-to-right pixel correspondences are computed by DIS optical flow on the *full* image (with variational refinement), and the analysis ROI is then extracted from the resulting field rather than computing the flow on a cropped ROI; within the  $1448 \times 1448$  px analysis window shown in Figure 8, every pixel carries a valid correspondence (the full-image flow leaves no holes in the ROI). For each correspondence, the two rays from the calibrated model are intersected at their closest point (ray-gap triangulation).

The median ray gap is now  $0.4 \mu\text{m}$  to  $0.5 \mu\text{m}$  across the displayed CMO and Zernike reconstructions in Figure 8, and all five variants in Table 8 show similarly tight intersections, confirming that the two channels' rays intersect each other at the sub-micrometre level despite never having been jointly optimised on this specimen. The CMO 26p reconstructions are smoother than the Zernike rayfield reconstruction because the compact physical model imposes implicit spatial smoothness through its rigid-sub-pupil and affine-direction parameterisation, whereas the 57-parameter Zernike model has the flexibility to absorb DIS correspondence noise into high-frequency surface variations. Quantitative Z MAD and ray-gap values for the rayfield-initialised, unregularised BA, isotropic-prior BA, and Schur-prior BA variants are reported in Table 8. OpenCV stereo reconstruction was not available because the standard OpenCV calibration failed to converge on this microscope's calibration data.

The specimen comparison is intentionally presented in two steps, because the apparent disagreement is scientifically useful. First, the SE(3) alignment shows that both models reconstruct the same surface up to the CMO model's  $v \rightarrow Y$  sign convention (a Y-axis reflection) and a small ( $\sim 8.6^\circ$ ) rigid rotation.



**Figure 8.** Dense specimen reconstruction. Top row: left image, DIS optical-flow magnitude  $\|(U,V)\|$ , valid mask (full coverage—the full-image flow leaves no holes in the ROI). Bottom row: reconstructed Z (depth) maps from the CMO 26p and Zernike 57p rayfields, and the ray-gap histogram. Both rayfields produce nearly identical, tight ray gaps (median  $\sim 0.5 \mu\text{m}$ ), confirming rayfield coherence. Quantitative surface roughness and ray-gap values for all five model variants (rayfield-initialised through Schur-regularised) are reported in Table 8.



**Figure 9.** Rigid-gauge removal: Zernike vs CMO surface comparison. Top row: CMO and Zernike surfaces after plane-normalisation (relief only), followed by  $\Delta Z$  before and after plane normalisation. Bottom row:  $\Delta Z$  before SE(3) alignment (dominated by an affine ramp,  $R^2 = 0.997$ ), the fitted affine plane, and the residual histogram. After SE(3) alignment the median plane-normalised  $\Delta Z$  residual is 0.0070 mm; the corresponding median 3D point-to-point residual drops from 3.5 mm to 0.018 mm (a 99.5% reduction). The remaining relief difference is a depth-only scalar (relief-amplitude ratio  $Z_{\text{Zernike}}/Z_{\text{CMO}} = 1.42\times$ , i.e.  $s_z \approx 0.70$ ): the two reconstructions agree up to a single depth-scale factor, not a spatially arbitrary non-rigid discrepancy. Which reconstruction is closer to the true surface is settled by external profilometry in §4.9.

Second, after these gauge effects are removed, a residual depth-relief discrepancy remains: the Zernike reconstruction exhibits approximately 42 % larger relief amplitude than the CMO 26p model (standard deviation ratio  $1.42\times$ , Figure 9). To characterise whether the discrepancy is a uniform scale error, a non-rigid shape error, or a depth-only effect, we fit nested transformations from Zernike to CMO points on matched specimen correspondences. An SE(3)-only fit (no scale) yields a median residual of 0.008 mm, already confirming excellent geometric agreement. This 0.008 mm is the SE(3)-only fit residual on the 50 000 subsampled correspondences, distinct from the 0.018 mm dense-map median 3D point-to-point residual reported in Figure 9. Adding a global isotropic scale (Sim(3)) produces an optimal scale factor  $s = 1.00$ , ruling out a uniform 3D magnification—the difference is not a simple focal-length or pixel-scale error. However, an anisotropic scale model with independent  $(s_x, s_y, s_z)$  recovers  $s_x \approx s_y \approx 1.00$  but  $s_z \approx 0.70$ , indicating a **depth-only scale difference** of approximately 30 % (consistent with the  $1.42\times$  relief-amplitude ratio above). The lateral dimensions  $(x, y)$  are essentially identical between the two reconstructions; only the reconstructed depth range differs, by the scalar factor  $s_z \approx 0.70$  above.

This depth-only scale difference is therefore not a non-rigid disagreement and not a contradiction of the CMO family identification: the two reconstructions agree up to a single depth-scale factor. The CMO model’s rigid sub-pupil and telecentric direction parameterisation impose a specific stereo geometry that the Zernike rayfield, being free, is not obliged to reproduce exactly.

Crucially, this residual axial freedom is a property of the calibration *pose set*, not of the Zernike rayfield. The Pycaso board poses are near-parallel planar positions: they constrain the lateral geometry strongly but the axial metric only weakly, because a global depth rescaling barely changes the reprojection of a fronto-parallel grid, leaving the fit little leverage to pin  $s_z$ . A calibration set that included substantially *tilted* board poses would make the axial scale observable—an axial rescaling distorts the reconstructed Euclidean square lengths once the board has a non-zero out-of-plane component, so the known grid spacing would then constrain  $s_z$  directly. The depth-scale ambiguity is therefore one of limited axial *excitation* by the available planar poses, not an inherent defect of the per-pixel ray-field. It is moreover immaterial to the method’s goal: the rayfield here is an *intermediate* measurement used to construct and identify a physical model, not the final calibrated product. Had the Zernike rayfield itself been the intended output, pinning its depth scale would have required either such tilted poses or an explicit axial-metric prior; for the model-construction role it plays here, neither is needed—the CMO model and its free-pose Schur-stabilised refinement supply the axial geometry that the planar pose set leaves loosely excited.

From the stereo data alone the depth-scale difference is a convergence-depth ambiguity inherent to uncalibrated stereo reconstruction rather than a failure of either model, and does not by itself establish which reconstruction is closer to the real coin surface. We therefore provide exactly the required independent axial reference—profilometry of the same specimen—in §4.9, which resolves the ambiguity in favour of the CMO 26p (and Soloff) reconstructions.

Table 8 quantifies the five reconstructions. All five variants—including the rayfield initialisation—recover tight ray intersections on the dense full-image correspondence field, with a median ray gap of  $\sim 0.5\ \mu\text{m}$ . The ray geometry is therefore correct regardless of the regularisation strategy, and the rayfield-identified initialisation already triangulates the specimen as tightly as the BA-refined models. The regularised variants achieve Z MAD  $\sim 0.028\ \text{mm}$ , comparable to the unregularised BA (0.023 mm)—the prior keeps the dense reconstruction comparable while preserving ray-gap and magnification consistency. On the dense full-image correspondence field all five variants converge to the same lateral magnification



( $\sim 0.199$  vs. the nominal coin): there is *no* systematic CMO–Zernike lateral bias. The apparent  $\sim 3.5\%$  bias seen with the sparser ULTRAFAST correspondences was an artefact of differing per-variant valid sets, not a property of the models. The lateral magnification, now consistent across variants, is in any case distinct from the depth-only scale difference ( $s_z \approx 0.70$ ) and cannot explain it: the specimen diagnostic separates lateral magnification from depth sensitivity.

**Table 8.** Five-variant specimen reconstruction: surface roughness (Z MAD), median ray gap, and magnification ratio vs. the nominal 18.75 mm 2-cent euro coin.

| Model                                                   | Z MAD (mm)   | Median gap (mm) | Magnification |
|---------------------------------------------------------|--------------|-----------------|---------------|
| Zernike rayfield (57 p)                                 | 0.202        | 0.0005          | 0.1991        |
| CMO 26 p (rayfield init)                                | 0.076        | 0.0004          | 0.1990        |
| CMO 26 p — BA unregularised                             | 0.023        | 0.0005          | 0.1979        |
| CMO 26 p — BA + isotropic prior ( $\alpha=10^{-2}$ )    | 0.028        | 0.0004          | 0.1985        |
| CMO 26 p — BA + <b>Schur prior</b> ( $\alpha=10^{-3}$ ) | <b>0.028</b> | 0.0004          | 0.1985        |

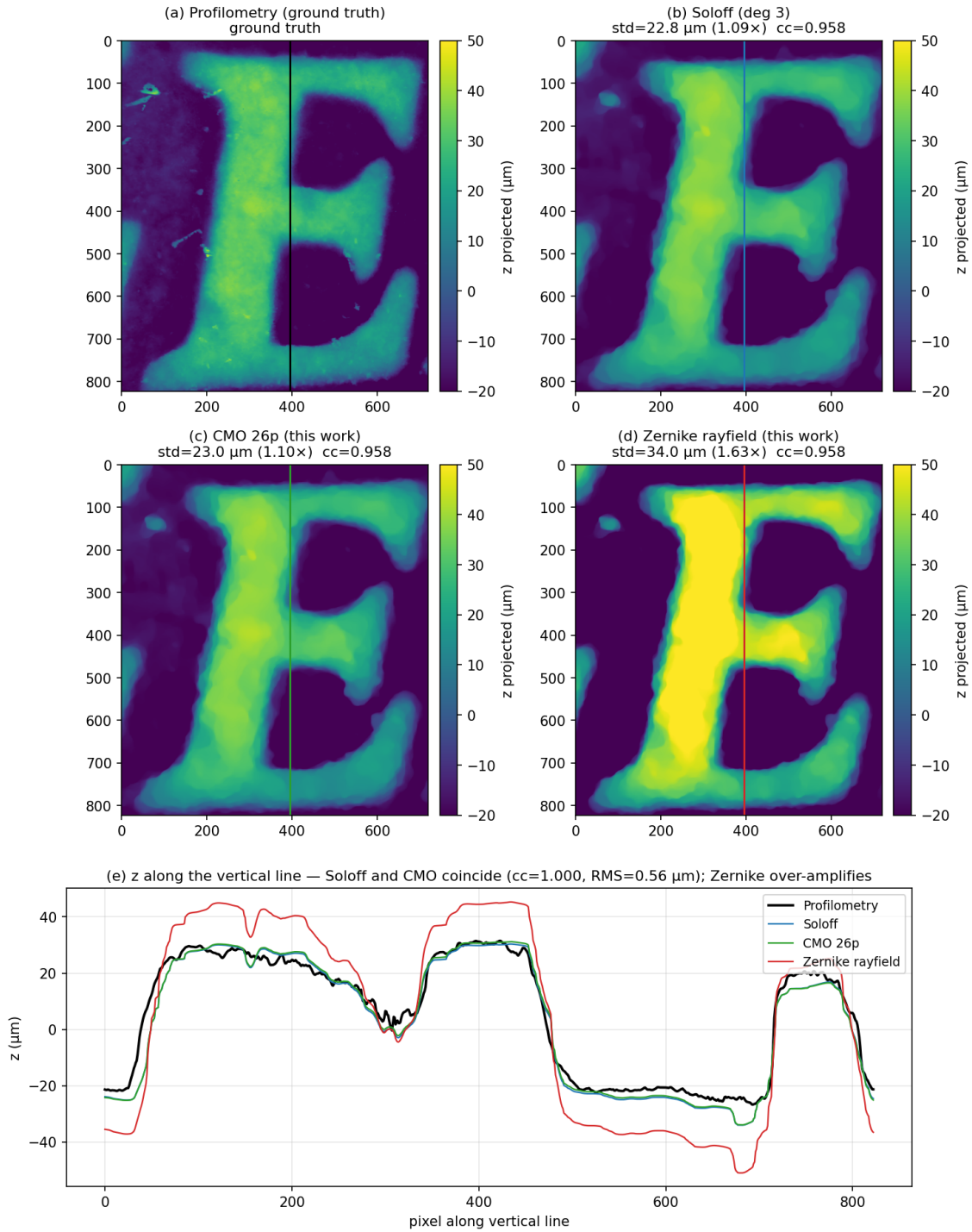
#### 4.9 External Profilometry Validation

The depth-scale ambiguity of §4.8 is, by construction, unresolvable from the stereo data alone: it asks which of two self-consistent reconstructions is closer to the real surface. We resolve it here with the independent axial reference the analysis called for—a profilometric scan of the same coin. The scan covers the embossed letter “E” and is registered to the left-camera pixel grid by a rigid similarity (scale  $\approx 1.03$ ) fitted on the binary letter mask (intersection-over-union 0.88), a flow-independent alignment that fixes the “E” geometry. Dense correspondences are recomputed with a refined DIS flow (full image, variational refinement) and the *same* field is reconstructed by three models; each plane-normalised relief is compared, on the common “E” window, against the profilometry.

The result (Figure 10) resolves the ambiguity for this specimen and ROI. Against the profilometric relief standard deviation of  $20.8\ \mu\text{m}$ , the Soloff polynomial recovers  $22.8\ \mu\text{m}$  ( $1.09\times$ ), the CMO 26p model  $23.0\ \mu\text{m}$  ( $1.10\times$ ), and the Zernike rayfield  $34.0\ \mu\text{m}$  ( $1.63\times$ ); all three correlate with the ground-truth shape at  $cc = 0.96$ . The CMO 26p and Soloff reconstructions are statistically *indistinguishable* on the relief (mutual  $cc = 0.9997$ , RMS difference  $0.57\ \mu\text{m}$ , an order of magnitude below their  $\sim 2\ \mu\text{m}$  distance to the profilometry). The Zernike-to-CMO amplitude ratio measured here,  $1.63/1.10 \approx 1.48\times$ , is consistent with the internal  $1.42\times$  relief ratio of Figure 9.

Two conclusions follow. First, the depth-only scale difference ( $s_z \approx 0.70$ ) is an *over-amplification by the per-pixel Zernike dense reconstruction*, not a loss of axial scale by the CMO model: the compact 26-parameter physical model’s relief amplitude agrees with the external profilometric reference within  $\approx 10\%$ , as does an unconstrained Soloff polynomial fitted to the same data. The modest  $\sim 10\%$  residual excess of both faithful models over the profilometry is consistent with dense-stereo (DIS) correspondence noise inflating the standard deviation (the profilometer is far cleaner) and, in part, with the residual imperfection of the lateral registration itself (mask IoU = 0.88, point-wise  $cc = 0.96$ , not unity); the  $\sim 10\%$  should therefore be read as an *upper bound* on the model’s intrinsic relief error rather than a calibrated metrological uncertainty. Second, the over-amplification is specific to the high-dimensional, non-central rayfield reconstruction—which has substantially more freedom than the compact 26-parameter CMO model—whose extra latitude converts local correspondence geometry into amplified depth; it is not a property of the Zernike basis nor of non-parametric calibration in general. The Zernike rayfield is

## External profilometry validation of the specimen relief



**Figure 10.** External profilometry validation of the specimen relief. Maps (a–d): plane-normalised relief of the embossed “E” from the profilometric ground truth, the Soloff polynomial, the CMO 26p model, and the Zernike rayfield, on a common registered window (relief standard deviation and amplitude ratio to the profilometry annotated). Panel (e): the depth profile along the vertical line—the profilometry (black), Soloff (blue) and CMO (green) coincide, while the Zernike rayfield (red) over-amplifies every plateau and valley. All three reconstructions track the ground-truth “E” shape equally well ( $\text{cc} \approx 0.96$ ); they differ only in axial gain.



**Table 9.** Robustness of the relief comparison to the dense-matching front-end. Amplitude ratios are the plane-normalised relief standard deviation relative to the profilometry; the last two columns quantify the CMO–Soloff agreement. The production DIS parameters were tuned against the profilometry, but the bare OpenCV preset defaults (no manual variational refinement) reproduce the same relative result:  $\text{CMO} \approx \text{Soloff}$  and the Zernike ray-field over-amplifies. The external comparison is therefore a consistency check of the calibrated models on a fixed correspondence field, not a tuning of the DIS front-end.

| DIS setting          | Soloff<br>( $\times$ ) | CMO 26p<br>( $\times$ ) | Zernike<br>( $\times$ ) | CMO–Soloff<br>RMS ( $\mu\text{m}$ ) | CMO–Soloff<br>cc |
|----------------------|------------------------|-------------------------|-------------------------|-------------------------------------|------------------|
| Tuned (production)   | 1.09                   | 1.10                    | 1.63                    | 0.56                                | 0.9997           |
| ULTRAFast (defaults) | 1.09                   | 1.10                    | 1.63                    | 0.58                                | 0.9997           |
| FAST (defaults)      | 1.06                   | 1.07                    | 1.59                    | 0.56                                | 0.9997           |
| MEDIUM (defaults)    | 1.12                   | 1.12                    | 1.67                    | 0.58                                | 0.9997           |

therefore a ray-space diagnostic reference, not a metrological ground truth for dense surface relief. This single-specimen comparison does not certify absolute accuracy—it is one specimen, and the profilometer sets the absolute scale—but it does support the CMO/Soloff depth scale on this specimen, against external data.

The production DIS parameters were selected after inspecting the profilometric agreement; the comparison is therefore an external *consistency check of the calibrated models on a fixed dense correspondence field*, not a blind validation of the DIS front-end. To confirm that this tuning does not manufacture the result, Table 9 repeats the entire comparison under the bare OpenCV ULTRAFast / FAST / MEDIUM preset defaults (no manual variational refinement): across every setting the CMO and Soloff reliefs stay indistinguishable (CMO–Soloff RMS 0.56  $\mu\text{m}$  to 0.58  $\mu\text{m}$ ,  $\text{cc} = 0.9997$ ) and the Zernike rayfield keeps over-amplifying (1.59–1.67 $\times$ ). The *relative* conclusion— $\text{CMO} \approx \text{Soloff}$ , Zernike over-amplifies—is thus invariant to the dense-matching front-end; the tuning shifts only the absolute amplitude ratios, by at most  $\pm 0.03$ .

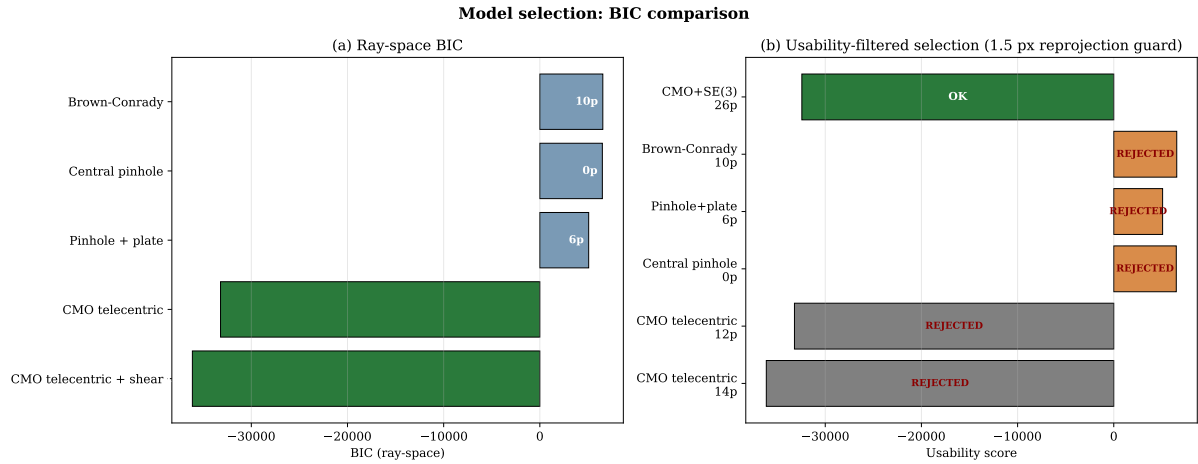
The Soloff polynomial is therefore used here as a strong black-box metrological baseline, not as an optical diagnostic. Even when it reconstructs the coin relief accurately, its polynomial coefficients do not define per-pixel rays, Plücker residuals, baselines, convergence angles, or Schur-observable optical modes. It validates surface reconstruction but cannot replace the rayfield-to-CMO identification step.

#### 4.10 Model Performance

**Table 10.** Model comparison on Pycaso CMO data. The 26-parameter CMO+SE(3) model is the compact physical reference. All rows use the operational constrained-pose protocol; the free-pose Schur-stabilised BA refines the same 26-parameter model to 0.28 px (Table 2), a diagnostic setting not listed here.

| Model                   | $p$       | Ray RMS (mm)  | Px RMS (px) | P50 (px)    | P95 (px)    |
|-------------------------|-----------|---------------|-------------|-------------|-------------|
| OpenCV stereo           | –         | –             | > 300       | –           | –           |
| Perspective CMO         | 19        | 3.48          | ~86         | –           | –           |
| Telecentric CMO + shear | 14        | 0.12          | 14.6        | 13.2        | 22.4        |
| <b>CMO + SE(3)</b>      | <b>26</b> | <b>0.0021</b> | <b>1.06</b> | <b>0.87</b> | <b>1.84</b> |
| CMO + SE(3) + corner BA | 26        | ~0.0019       | 0.88        | 0.64        | 1.70        |
| Zernike O(0)+d(2)       | 57        | 0.0007        | 0.47        | 0.34        | 0.86        |

The 26-parameter CMO+SE(3) model achieves 1.06 px reprojection with less than half the parameters of the Zernike reference. It has a physically interpretable parameterisation: sub-pupils, effective axial length, telecentric slopes, and SE(3) arm corrections.



**Figure 11.** Two-panel BIC bar chart. (a) Ray-space BIC: the CMO telecentric family achieves the lowest BIC by a decisive margin ( $> 40\,000$  points). (b) Operational usability score with a 1.5 px reprojection guard: models exceeding the threshold are marked as REJECTED (shown in grey/orange); only the 26p SE(3)-aligned model passes the guard (shown in green).

#### 4.11 BIC Model Selection

**Table 11.** BIC model selection on the Zernike rayfield. Models above 1.5 px reprojection are REJECTED by the operational guard. The central and parallel-plate models have multi-millimetre ray RMS and are rejected on reprojection alone; their observability-penalised score  $S_{\text{usable}}$  then diverges ( $\gg 0$ ) and is not tabulated precisely.

| Model                  | $p$       | RMS (mm)     | $\text{BIC}_{\text{ray}}$ | $S_{\text{usable}}$ | Status             |
|------------------------|-----------|--------------|---------------------------|---------------------|--------------------|
| cmo telecentric shear  | 14        | 0.111        | -36 128                   | +978 890            | REJECTED           |
| cmo telecentric        | 12        | 0.146        | -33 201                   | +986 044            | REJECTED           |
| pinhole parallel plate | 6         | 6.337        | 5 079                     | $\gg 0$             | REJECTED           |
| central pinhole        | 0         | 8.343        | 6 500                     | $\gg 0$             | REJECTED           |
| central brown conrady  | 10        | 8.324        | 6 549                     | $\gg 0$             | REJECTED           |
| <b>CMO + SE(3) 26p</b> | <b>26</b> | <b>0.002</b> | <b>-32 433</b>            | <b>-32 433</b>      | <b>BEST USABLE</b> |

The ray-space BIC provides decisive rejection of the non-CMO families ( $\Delta\text{BIC} > 40\,000$ ), confirming that pinhole, Brown-Conrady, and parallel-plate models are grossly misspecified for this non-central system. The BIC magnitudes should be interpreted as rejection of families, not as certification of the retained model: the rayfield residuals are spatially correlated (the Zernike field is smooth), so the effective number of independent observations is lower than  $N_{\text{obs}} = 3300$ , inflating the absolute BIC scale. The ranking across families is robust to this effect. The operational usability score correctly rejects the 14p base model as unusable and selects the 26p aligned model.

#### 4.12 Geometric Descriptors

Table 12 collects the CMO-consistent effective geometric descriptors read directly from the centre-pixel rayfield under the chosen gauge.

**Table 12.** CMO-consistent effective geometric descriptors read directly from the Zernike rayfield at the centre pixel. All values are reported in the chosen rayfield gauge ( $O \cdot d = 0$ ) and fixed reference focal scale ( $f_x = 25600$  px); their absolute metrological interpretation requires an independent external reference (see Section 5.2).

| Descriptor             | Symbol           | Value   |
|------------------------|------------------|---------|
| Stereo baseline        | $b$              | 24.9 mm |
| Sub-pupil depth        | $z_p$            | 2.5 mm  |
| Working distance       | $WD$             | 64.7 mm |
| Effective axial length | $f_{\text{obj}}$ | 62.2 mm |
| Convergence angle      | $\theta$         | 22.6°   |

#### 4.13 SE(3) Ablation

**Table 13.** SE(3) arm alignment ablation. Per-arm DOFs are individually necessary.

| Variant              | Params    | Ray RMS (mm)  | Px RMS (px) | P50 (px)    | P95 (px)    |
|----------------------|-----------|---------------|-------------|-------------|-------------|
| Telecentric baseline | 14        | 0.12          | 14.6        | 13.2        | 22.4        |
| + Rotation only      | 20        | 0.014         | 3.74        | 2.68        | 7.13        |
| + Translation only   | 20        | 0.010         | 2.44        | 2.06        | 4.15        |
| <b>+ Full SE(3)</b>  | <b>26</b> | <b>0.0021</b> | <b>1.06</b> | <b>0.87</b> | <b>1.84</b> |
| + Shared rotation    | 23        | 0.0083        | —           | —           | —           |
| + Shared translation | 23        | 0.0041        | —           | —           | —           |
| + Differential only  | 20        | 0.0070        | 4.04        | 2.85        | 7.74        |

Both rotation and translation are essential; neither alone suffices (Table 13). Per-arm DOFs are individually necessary—any compression degrades the fit (+92 % to +289 % ray RMS increase).

## 5 Discussion

### 5.1 The Rayfield as Diagnostic Instrument

The key methodological contribution is the feedback loop: Ray2D preprocessing → Ray3D measurement → residual modal analysis → hypothesis → model refinement → verification. Without the rayfield, the 2D reprojection error tells you *that* the model is wrong but not *how*. The Zernike projection of  $\Delta d$  and  $\Delta m$  directly identifies whether the missing DOF is global (low-order Zernike modes) or spatial (high-order modes).

Figure 5 illustrates this diagnostic value concretely. The residual heatmap at each model stage is a **visual hypothesis generator**: the perspective CMO’s structured  $d_y$  error (23.6° RMS) sends the user to telecentric models; the telecentric model’s uniform  $Z_0^0$  piston (0.27° RMS) sends the user to global arm alignment; and the SE(3)-refined model’s near-zero residual (0.003° RMS) confirms convergence. At no point does the user need to guess—the residual pattern at each stage dictates the next hypothesis.

In this case study, the  $Z_0^0$ -dominated residual correctly guided us to the SE(3) arm alignment, while alternative hypotheses (pre-warp, variable origin) were correctly rejected by the same diagnostic.

### 5.2 Limitations

1. **Single dataset.** Results are demonstrated on 10 stereo pairs from one Pycaso CMO microscope. Generalisation to other CMO designs, manufacturers, or magnification settings has not been

demonstrated and requires additional validation.

2. **Calibration residuals use the board points.** All ray-space and pixel-equivalent residuals are computed on the same ChArUco board points used for calibration. The independent surface check is profilometry of the specimen (§4.9); a separate *certified* 3D artefact at a known distance remains unavailable for the open Pycaso dataset.
3. **Depth-scale ambiguity (resolved by profilometry).** The specimen sanity check (§4.8) reveals a  $\sim 30\%$  depth-only scale difference between the Zernike and CMO 26p reconstructions ( $s_z \approx 0.70$ ). External profilometry of the same specimen (§4.9) resolves it: the CMO 26p relief matches the profilometric ground truth to  $\approx 10\%$ —as does a Soloff polynomial fitted to the same data—whereas the Zernike-based dense reconstruction over-amplifies the relief by  $1.63\times$ . The depth-scale gap is therefore Zernike over-amplification, not a CMO depth loss. Given its point-wise lateral registration ( $cc = 0.96$ ) and the profilometer-set scale, this single-specimen comparison supports the CMO/Soloff depth scale but does not by itself certify an absolute depth uncertainty.
4. **Effective rayfield descriptors are gauge-sensitive.** The geometric descriptors ( $b, WD, f_{\text{obj}}, \theta$ ) are read from the rayfield under the transverse gauge  $O \cdot d = 0$  and a fixed pinhole reference  $f_x = 25600$ . The working distance is linearly proportional to the initial  $f_x$  ( $\pm 10\% f_x \rightarrow \pm 10\% WD$ , Section 4.7), and is *not* independently identifiable without an external scale reference.
5. **SE(3) translation parameters are not uniquely identifiable.** The rotation angles ( $\sim 2.5^\circ, \sim 3.7^\circ$ ) are stable across optimisation runs, but the translation components trade off with the telecentric base parameters ( $WD, f_{\text{obj}}, b$ , principal point). Only the SE(3) rotation is a robust diagnostic; the translation is effective rather than absolute.
6. **The Zernike rayfield is a training residual.** Its  $0.0007\text{ mm}$  ray-space RMS is a self-evaluation on the support points it was fitted to; the  $0.47\text{ px}$  pixel RMS is a noise-floor estimate, not an independent measurement. The pixel RMS column in all tables provides the apples-to-apples comparison.
7. **Fixed pinhole reference.** The Zernike BA uses a fixed  $f_x = 25600$  and principal point at image centre. Errors in this reference propagate to the rayfield estimates and the derived physical descriptors.
8. **Monochromatic effective model.** The rayfield and all derived physical models are identified from broad-spectrum white-light images and represent an effective model at the sensor's integrated spectral band (visible,  $\sim 400\text{ nm}$  to  $700\text{ nm}$ ). CMO microscopes carry residual chromatic aberration—both lateral (magnification variation with wavelength) and longitudinal (focus shift)—that is absorbed into the effective ray directions and origins per pixel. A wavelength-resolved rayfield  $\mathcal{R}(u, v, \lambda)$  is a natural extension but requires narrow-band or hyperspectral acquisitions not available in the present dataset; it would disentangle geometric misalignment from chromatic effects and is noted as a future direction.
9. **Absolute metrological validation would require:** (a) manufacturer specifications for the actual microscope used, (b) an independent measurement of WD or baseline, or (c) a certified 3D

calibration object at a known distance. Item (a) is now partly available—the instrument is identified as a Leica M205C with a Planapo 1.0 $\times$  objective (documented  $WD = 61.5$  mm [13]), and the reconstructed baseline agrees with it to  $\approx 1\%$  via the chief-ray relation (15) (§4.5)—but this is a physical *consistency* check that still uses the datasheet WD as the scale anchor, not an independent measurement; items (b) and (c) remain unavailable for the open Pycaso dataset. The present validation thus establishes internal geometric consistency, agreement with the documented optical configuration, model interpretability, and descriptor stability on a real CMO dataset, but not absolute 3D metrological accuracy. A future validation on a calibrated 3D artefact or stepped depth standard will be required to quantify absolute depth accuracy and uncertainty propagation.

### 5.3 Generalisability

The methodology—measure with a flexible non-parametric basis, read physical descriptors, diagnose residual structure, build compact physical model, validate by BIC—is transferable to any non-standard optical instrument (endoscopes, Scheimpflug cameras, multi-camera arrays). The SE(3) arm alignment is specific to CMO microscopes, but the residual analysis that identified it is general.

### 5.4 Why the Two-Stage Decomposition Works

The central insight of this work is not a specific model architecture but a problem decomposition strategy. Direct bundle adjustment jointly estimates optical parameters  $\theta$  and board poses  $\eta$  from 2D corners; for non-central optics, the Schur complement of the reduced camera system [33] reveals coupling norms of  $c = 0.81$ – $0.98$  depending on the diagnostic configuration, meaning optics and poses are nearly inseparable—the problem is severely ill-conditioned. By inserting the Zernike rayfield as an intermediate variable, the estimation splits into two sub-problems: (1) fit a flexible rayfield to corners with a constrained pose model, and (2) identify physical optics in ray space, where the coupling norm is 0.0 by construction because board poses are no longer estimated (pose-free model selection with BIC).

This decomposition is the mechanism behind all three empirical findings: the TPS pre-processing stabilises step (1) by cleaning the 2D input; the rayfield decouples step (2) from pose estimation; and the rayfield-initialised parameters lie close to the constrained-pose corner optimum, while also providing the correct basin and observability prior for the free-pose Schur-stabilised refinement.

## 6 Conclusion

We have demonstrated the first real-data validation of the StereoComplex non-central calibration pipeline on a CMO stereo microscope. The key results are:

- A compact 26-parameter physical model (quasi-telecentric CMO skeleton + per-channel SE(3) arm alignment) achieves 1.06 px reprojection (P50=0.87 px, P95=1.84 px) where standard OpenCV stereo calibration fails ( $> 300$  px).
- The Zernike rayfield serves as a diagnostic instrument: residual modal analysis identifies missing degrees of freedom, and the operational usability score with reprojection guard correctly selects the usable model.
- Effective rayfield descriptors—baseline, working distance, effective axial length, convergence angle—are read directly from the measured rays (within the chosen rayfield gauge and reference scale), without fitting a physical model.

- On the independent coin specimen, the remaining CMO–Zernike relief discrepancy is a single depth-only scale factor ( $s_z \approx 0.70$ ), not a non-rigid error; external profilometry of the specimen (§4.9) resolves it in favour of the CMO model, whose relief matches the profilometric ground truth to  $\approx 10\%$ —as does a Soloff polynomial—while the Zernike-based dense reconstruction over-amplifies the relief by  $1.63\times$ .
- The rayfield-initialised parameters are near-optimal for direct corner bundle adjustment (17% improvement), and the same rayfield provides a Schur-complement observability prior that stabilises the BA against optics–pose compensation.
- The accuracy–interpretability trade-off is quantified by comparison with the Soloff polynomial baseline (§4.6): a black-box polynomial achieves 0.18 px (80 params) but yields no physical descriptors; the Zernike rayfield reaches 0.41 px (93 params) under the 3D geometric consistency constraint (0.23 px gap = cost of geometry); the compact CMO 26p model is identified at 1.06 px and, used as the initialisation for the free-pose Schur-stabilised bundle adjustment, refines to 0.28 px (within 0.1 px of the Soloff floor) with a physically interpretable parameterisation. We therefore do not merely trade accuracy for interpretability: the refined physical model is competitive in pixel accuracy with black-box regression while, unlike it, producing a physically meaningful calibration.

The CMO Pycaso case study validates a generalisable methodology—measure first, model second, diagnose by residual, refine by hypothesis testing—and establishes the Zernike rayfield as a multi-purpose instrument: it measures the rays, diagnoses the missing physical degree of freedom, initialises the bundle adjustment, and regularises it in two distinct senses—*structurally*, its 3D ray constraint resists overfitting and extrapolates better than an unconstrained polynomial (§4.6), and *statistically*, its Schur-complement spectrum supplies an observability prior that suppresses optics–pose compensation (§3.9). Notably, the axial-scale ambiguity left by the planar pose set (§4.8) never impeded this construction: the compact physical model was initialised, built, and—through the free-pose Schur-stabilised bundle adjustment—identified to 0.28 px regardless, precisely because the rayfield serves here as a model-construction instrument rather than a metrically self-sufficient calibration end-product. Code, data, and documentation are publicly available at <https://github.com/jeffwitz/StereoComplex>. A self-contained Zenodo archive (concept DOI [10.5281/zenodo.20444215](https://doi.org/10.5281/zenodo.20444215)) bundles the manuscript, figures, tables, numerical audit, reproducibility scripts, and all data files needed to rebuild the PDF and regenerate every figure. For exact reproducibility, the version submitted with this manuscript is archived as v6 ([10.5281/zenodo.20575213](https://doi.org/10.5281/zenodo.20575213)); the concept DOI above is a stable landing page that always resolves to the latest version.

## Prospective

Several directions extend this work. First, the methodology should be validated on additional CMO microscope designs and manufacturers to confirm generalisability beyond the single Pycaso instrument. Second, the Greenough stereo microscope family—with tilted optical axes rather than parallel sub-pupils—presents a harder identification problem where the telecentric model does not apply; the residual-guided construction method would need to discover a different geometric skeleton. Third, quantitative uncertainty propagation from ChArUco corner noise to effective-descriptor estimates (baseline, working distance) would strengthen the metrological claim. Fourth, deep learning-based corner detection could replace the hand-tuned Hessian+TPS pipeline, potentially improving the raw 2D input quality and further reducing



the gauge instability. Finally, extending the Zernike rayfield to time-varying deformations (e.g., vibration, thermal drift) would enable *in-situ* calibration monitoring without dedicated targets.

## Reproducibility Statement

The StereoComplex codebase (commit f912331, branch `develop`) and all artefacts required to reproduce these results are available at <https://github.com/jeffwitz/StereoComplex>. The raw Pycaso calibration images are available from <https://github.com/LaboratoireMecaniqueLille/Pycaso>. To reproduce the numerical results without access to the raw images, restart from the pre-computed intermediate state (`intermediate_state.npz`) which contains the already-detected, Hessian-completed, and TPS-denoised corner positions, the fitted Zernike rayfield, and the initial 26p model parameters. A self-contained Zenodo archive provides the manuscript, all figures, tables, numerical audit, reproducibility scripts, figure-generation manifests, and key data files (`intermediate_state.npz`, specimen correspondences, Schur diagnostic). For exact reproducibility, the version submitted with this manuscript is archived as v6 ([10.5281/zenodo.20575213](https://doi.org/10.5281/zenodo.20575213), a structured self-contained bundle—package source, reproduction scripts and key data, preserving the repository layout—plus the manuscript PDF); the concept DOI ([10.5281/zenodo.20444215](https://doi.org/10.5281/zenodo.20444215)) is a stable landing page under which future versions are deposited.

## A Numerical Reprojection for the CMO Model

The CMO telecentric model defines a forward mapping  $\text{ray}(u, v) \rightarrow (O, d)$  from pixel coordinates to a 3D ray. Computing the 2D reprojection error requires the inverse: given a 3D point  $X^{\text{cam}}$  in the camera frame, find the pixel  $(u, v)$  whose ray passes through (or closest to)  $X^{\text{cam}}$ . For a non-central model, no closed-form inverse exists. We solve it numerically: for each observation, the predicted pixel is

$$(u_{\text{pred}}, v_{\text{pred}}) = \underset{(u,v) \in [0,W] \times [0,H]}{\text{argmin}} \|X^{\text{cam}} - P(u, v; \theta)\|^2, \quad (17)$$

where  $P(u, v; \theta)$  is the point on the ray  $(O(u, v), d(u, v))$  closest to  $X^{\text{cam}}$ , computed as

$$P = O + ((X^{\text{cam}} - O) \cdot d) d. \quad (18)$$

The objective is the squared point-to-ray perpendicular distance. The minimisation is performed with the L-BFGS-B algorithm [6], starting from the observed pixel  $(u_{\text{obs}}, v_{\text{obs}})$ , with bounds  $[0, W-1] \times [0, H-1]$ . Convergence tolerance is set to `ftol`= $10^{-12}$ , `gtol`= $10^{-8}$ . For the Pycaso dataset (10 frames, 165 corners, 2 channels; 3300 observations), the full reprojection computation takes approximately 35 seconds on a single CPU core.

The 2D reprojection residual for an observation is the vector  $(u_{\text{pred}} - u_{\text{obs}}, v_{\text{pred}} - v_{\text{obs}})$  in pixels. The 2D RMS reported in Section 3.9 and Table 2 is

$$\text{RMS}_{2D} = \sqrt{\frac{1}{N} \sum_{k=1}^N [(u_{\text{pred},k} - u_{\text{obs},k})^2 + (v_{\text{pred},k} - v_{\text{obs},k})^2]}, \quad (19)$$

where  $N = n_{\text{frames}} \times n_{\text{corners}} \times 2$  is the total number of observations.

This numerical reprojection is used as a post-optimisation diagnostic: the bundle adjustment itself minimises the ray-space residual (:func:`point\_to\_ray\_residuals\_cmo\_se3` in the StereoComplex

codebase), which is a forward evaluation and therefore faster and smoother. After convergence, the 2D reprojection error is computed from the optimised parameters to obtain the pixel-equivalent metric reported in the paper.

### Author Contributions

J.-F. Witz conceived the study, developed the StereoComplex framework, implemented all methods, conducted the experiments, and wrote the manuscript.

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### Code and Data Availability

- StereoComplex: <https://github.com/jeffwitz/StereoComplex> (Code: GPL-2.0-or-later. Documentation: CC BY-SA 4.0.)
- Pycaso dataset: <https://github.com/LaboratoireMecaniqueLille/Pycaso>
- All generated artefacts: docs/assets/pycaso\_real\_data/
- Intermediate state (restart without raw images):  
docs/assets/pycaso\_real\_data/intermediate\_state.npz

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